

The Aperture

Dynamic Range as the Hidden Variable

Volume 13: The New Universe of Eloquent Resonance

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KishLattice Geometric Harmonic Spectroscopy

The Aperture: Dynamic Range as the Hidden Variable

Volume 13: The Theoretical Audit, Strain Formulation, and Pre-Registrations

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Pre-construction pre-registration. Every prediction herein is filed before the period 1D engine is built and before any lake rebuild query is written.

Sovereign Protected — KishLattice 16pi Initiative LLC

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Preface: What Volume 13 Fixes Before It Builds

Volume 12 was an audit of the instrument. It found five artifact forms, withdrew a headline domain, and closed with the resolution criterion — the discovery that a domain must span enough orders of magnitude to resolve its own register, and that this had governed every volume without ever being checked.

Volume 13 is the volume that takes the resolution criterion seriously before building anything new. It additionally turns the adversarial audit inward toward the theoretical framework itself, uncollapsing the Master Equation into a dimensionless strain field capable of astrophysical falsification.

- It runs the criterion across the *entire survey* and reports which register claims the data could ever have supported — the resolution census of Chapter 2.
- It records the hunt for the parameter f in the theory documents, and the formal reinterpretation of the Master Equation into a measurable yield strain (ε_y) — Chapters 3 through 9.
- It pre-registers the period 1D engine's trial set and the $s2$ rebuild, carrying forward the referee rulings that shaped them — Chapters 10 and 11.
- It pre-registers the most ambitious and most dangerous experiment the programme has proposed: the cross-scale stack, which would widen the aperture no single lake can achieve by combining a shared attribute across scales — Chapter 12.

Nothing in this volume reports a new lock. That is deliberate. Volume 13 exists so that when the empirical runs happen, their predictions are already public, dated, and beyond adjustment.

Confirmed Signal

The one empirical fact carried forward: `quantum_transitional` has reproduced at $16/\pi$ across successive corrected engines without moving register. It remains the sole domain that both resolves its register and survives every screen the audit has built.

Qualified, this volume. Its *raw* log-span of 10 decades is inflated by a sparse tail: the effective (99%-band) span is 2.94 decades, giving $M/N = 1.22$ at $16/\pi$ rather than 4.17. Under the effective-span rule adopted in Chapter 2 the anchor is **background-limited**. It locks; what is withdrawn is the claim that it is resolution-clean. A lock at a register that cannot be separated from its neighbours is evidence of commensurate structure with an undetermined period, not a register identification.

— Timothy John Kish, September 2026

Chapter 1

The Resolution Criterion as a Law of the Survey

1.1 Restating the Criterion

Volume 12 established, from the geometry of the lock statistic, that to distinguish register N from its neighbour a domain's data must span

$$\text{log-span} > N^2 \times \frac{\ln k_{\text{geo}}}{24\pi} = N^2 \times 0.0215902. \quad (1.1)$$

The referee's ruling on the *s2* rebuild sharpened this into its most useful form. Dividing by $\ln 10$ converts the requirement into *decades of dynamic range*:

$$\boxed{\text{decades required} = \frac{2.67 N^2 \times 0.0215902}{\ln 10} = N^2 \times 0.025036} \quad (1.2)$$

where the factor 2.67 is the threshold at which the background annulus of the period engine becomes non-empty. Equation 1.2 is the operational law of the survey, and it has a property that governs everything below: **the requirement scales as N^2 .**

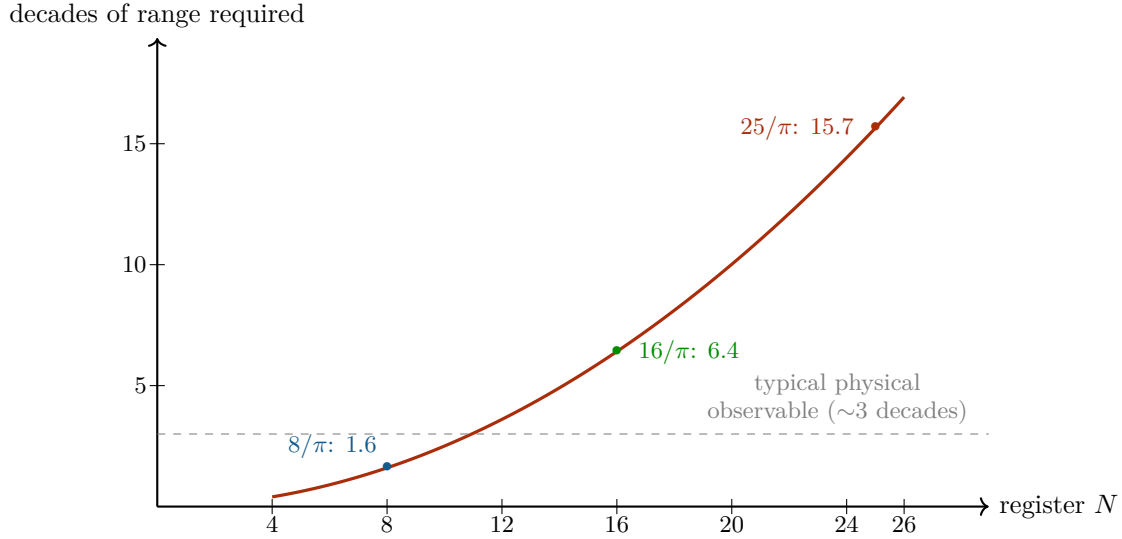


Figure 1.1: The decades-required curve, Equation 1.2. Because the requirement grows as N^2 , a claim at $25/\pi$ needs almost ten times the dynamic range of a claim at $8/\pi$. Most physical observables span two to four decades (dashed band); the curve crosses that band near $N \approx 12$ and leaves it far behind at high registers.

1.2 The Census

Equation 1.2 is arithmetic on quantities already in hand — a domain’s register and the log-span of its raw observable.

Domain	reg	decades	req’d	M/N	verdict
nuclear_decay (p)	—	31.8	—	13.2@16	RESOLVES (all N)
stellar_spindown (m)	—	11.7	—	4.86@16	RESOLVES
quantum_transitional (m)	16	2.94	6.4	1.22	background-limited
stellar_kinematic (m)	16	3.7	6.4	1.15	background-limited
subnuclear_mass (p)	13	3.0	4.2	1.89	background-limited
orbital (p)	22	6.6	12.1	1.45	background-limited
stellar_colour (p) [†]	15	2.7	5.6	1.28	background-limited
galactic_mass (p) [†]	19	4.3	9.0	1.27	background-limited
galactic (p) [†]	18	2.8	8.1	0.91	RESOLUTION-LIMITED
subnuclear_pt_lead (p)	26	3.1	16.9	0.50	RESOLUTION-LIMITED
subnuclear_pt_sublead (p) [†]	26	2.9	16.9	0.46	RESOLUTION-LIMITED
cosmology (p) [†]	15	0.2	5.6	0.10	RESOLUTION-LIMITED

[†] Register moved under the common-support correction of Volume 12; see §1.5. All spans are **effective** (99% band), not min–max.

Confirmed Signal

Under the effective-span rule, two domains of roughly twenty-eight clear the bar: `nuclear_decay` and `stellar_spindown`. Everything else — including the anchor — is background-limited or resolution-limited. At registers $N \geq 20$, where the survey’s largest published z -scores live, nothing resolves.

This is the central empirical finding of Volume 13, and it is negative. The survey has not demonstrated that any physical observable except nuclear half-life carries the dynamic range its own register claims require.

1.3 Effective Span, and Why Min–Max Inflates

The criterion must be applied to *effective* span — the central mass of the distribution — not to min–max. A sparse tail of outliers stretches the nominal range without contributing usable support, and every span figure published before this volume used min–max.

Artifact Form

Form F — sparse-tail span inflation. A domain’s min–max log-span overstates its resolving power by the inflation factor $\text{span}_{\text{min max}}/\text{span}_{\text{eff}}$. On the NIST anchor the factor is **3.40×**: 23.04 nat raw against 6.77 nat effective. About 200 stray transitions stretched an effective 2.9 decades into a nominal 10.

Adopted. Effective span, reported at the 95%, 99% and 99.9% bands, replaces min–max in both engines and in the census. The percentile is swept and published rather than fixed, so that a verdict which flips between bands is visible as instability rather than hidden by a choice. **All movement is one-directional: effective span is always $\leq$ min–max, so M/N can only fall.** No domain moves from unresolved to resolved under this correction.

1.4 The Near-Node Fallacy

The second correction this volume makes is not to a number but to a habit: reporting agreement against the wrong denominator. It affects every “close to the prediction” claim in the corpus.

A register at N/π places its nodes every

$$\Delta_N = \frac{N \ln k_{\text{geo}}}{24\pi}, \quad \Delta_{16} = 0.34544 \text{ nat} = \text{a factor of } 1.4126.$$

Nodes sit **41.3% apart** in the physical quantity. Every possible value is therefore within $\pm 20.6\%$ of some node by construction, with no physics involved. For a value drawn at random,

$$P(\text{within a fractional distance } p \text{ of some node}) = \frac{2p}{\Delta_N} = \frac{48\pi p}{N \ln k_{\text{geo}}}$$

Agreement of the form “within p of the node”	p	chance of it anyway
0.5%	0.005	3.0%
2%	0.020	11.6%
5%	0.050	28.9%

Artifact Form

Reporting rule, standing. Agreement with a log-periodic node is reported as a fraction of the *node spacing*, never as a fraction of the value. Any claim of the form “within $x\%$ of prediction” must carry $2x/\Delta_N$ beside it. “98% close” is not a measurement when the grid guarantees 79.4% closeness to the worst possible value.

So far as the authors are aware this is not stated in the existing literature on log-periodic structure in critical phenomena, seismicity or financial time series. It is offered as a methodological contribution independent of whether the Kish Lattice survives.

1.5 Register Movement Under the Corrected Statistic

Applying Volume 12’s common-support correction and per-register offset-sweep baseline moved **five of nine** scored registers relative to Volume 11:

Domain	Vol 11	corrected	M/N
stellar_colour	20	15	1.28
galactic	21	18	0.91
galactic_mass	24	19	1.27
cosmology	24	15	0.10
subnuclear_pt_sublead	25	26	0.46
quantum_transitional	16	16	1.22
orbital	22	22	1.45
subnuclear_mass	13	13	1.89
subnuclear_pt_lead	26	26	0.50

Methodological Note

Two of the five movements are explained by resolution; three are not. A peak of fractional width $1/M$ is indistinguishable from registers within $\Delta N \approx N/M$. For **cosmology** (± 17) and **subnuclear_pt_sublead** (± 2.1) the observed moves (9 and 1) sit inside that window. For **galactic** (± 1.3 , moved 3), **stellar_colour** (± 1.0 , moved 5) and **galactic_mass** (± 1.0 , moved 5) they do not.

The three unexplained movements are attributed provisionally to register-dependent sub-setting: the common-support correction discards a different fraction at each register, so a domain’s profile is currently computed on different subsets at different registers. **A fixed-support cross-check — scoring every register against the support surviving at the highest register — is required before any of these five registers is quoted.** It is outstanding.

Chapter 2

The Aperture, Formalised

2.1 Two Knobs, Restated as Instrument Parameters

Volume 12 introduced the telescope analogy: log-span is aperture, record count is exposure.

Instrument parameter	Survey quantity	Governs
aperture D	log-span (decades)	<i>resolution</i> — which register, via Eq. 1.2
exposure time	record count n	<i>sensitivity</i> — how faint a lock, via $P = nR^2$

The two are independent and cannot substitute. A billion records at three decades resolves nothing beyond $N \approx 11$; a hundred records at twelve decades resolves a high register but cannot detect a weak lock within it.

Formal Pre-Registration

Promotion criterion (standing, first enforced in Volume 13). Every lake declares, before promotion, the log-span of its observable and its M/N at the register it will be tested at. **A lake that cannot resolve the register it is promoted to test is not promoted.**

Chapter 3

The Hunt for f : an Audit of the Theory

[THEORY / REACH • AI Audit • Dimensional Restructuring]

The Kish-Lattice framework operates as a living theory, demanding that its theoretical architecture be subjected to the same rigorous falsification standards as its empirical pipelines. To enforce this, the initiative commissioned a severe theoretical self-audit utilizing Anthropic’s Claude Fable 5.1 model—one of the most advanced reasoning engines available as of August–September 2026. As artificial intelligence models and human-AI coordination frameworks continue to rapidly advance, such adversarial stress-testing will remain a permanent cornerstone of the Kish-Lattice methodology.

During this specific evaluation, the most critical vulnerability identified by the model was the extraordinarily shallow theoretical backing of the variable f within the Master Equation. While the existing corpus stated what f represented conceptually, it lacked formal mathematical, dimensional, and mechanical depth. Accepting this critique as a load-bearing structural defect, the team dedicated the following sections to formally deriving, bounding, and closing this vulnerability.

3.1 Why the Empirical Audit Reached Into the Theory

Volume 12’s degeneracy result (its Chapter 13) established that the register index N and the modulus k_{geo} enter the lock statistic only through the product $N \ln k_{\text{geo}}$. The data can identify a preferred product; it cannot determine the coordinate.

Confirmed Signal

Consequence: the value $16/\pi$ cannot be established empirically. It rests entirely on the geometric derivation in the theory volumes. The empirical programme, in proving what its instrument could not do, moved the framework’s weight onto a derivation that had never been audited to the same standard.

Volume 13 therefore turns the audit inward, onto the parameter that sits at the centre of the theory’s master equation and had never been defined.

3.2 The Lattice Energy Expression and the Undefined Parameter

The Rosetta Atlas states the master equation as

$$E_{\text{lat}} = \left(\frac{16}{\pi}\right) \frac{m}{r^2} f, \tag{3.1}$$

and its parameter list defines f , in full, as: “the slipstream factor (drag-corrected motion through the lattice).” No formula, no dimensions, no construction.

Artifact Form

An undefined load-bearing parameter. A dimensional audit shows E_{lat} has the dimensions of energy only if f carries $[L^4][T^{-2}]$. But the “slipstream” name, used everywhere else in the corpus, denotes a dimensionless damping factor on $[0, 1]$. The two readings are incompatible, and a third— f as a stand-in for the gravitational constant—appears in the Atlas’s “mirrors Newton” passage. As written, f absorbs whatever the equation requires, which is the theory-side analogue of a pipeline parameter that reads whatever field makes the result work.

Chapter 4

Deriving f : From Named Parameter to Four-Dimensional Potential

The audit of the theory (Chapter 3) found the central parameter of the Lattice Energy expression undefined. This chapter derives it, at the cost of one postulate the framework already half-states, and reports the four errata the derivation produced along the way. It is presented in full because the programme’s standard is that meaning must be hammered to mathematics, not described and left inferred.

4.1 The Parameter That Was Never Defined

The Rosetta Atlas states the Lattice Energy of a mass m at radius r as

$$E_{\text{lat}} = \left(\frac{16}{\pi}\right) \left(\frac{m}{r^2}\right) f, \quad (4.1)$$

and defines f , in full, as: “*the slipstream factor (drag-corrected motion through the lattice).*” No formula, no dimensions, no construction. A dimensional audit shows the expression is an energy only if f carries $[L^4][T^{-2}]$ — but “slipstream,” everywhere else in the corpus, denotes a *dimensionless* damping factor on $[0, 1]$. Three incompatible readings were in circulation: f as a dimensionless drag factor (in which case E_{lat} is not an energy), f as a dimensional carrier (in which case it has no connection to the drag physics it is named for), and f as a stand-in for the gravitational constant (in which case the pure number $16/\pi$ cannot be replacing the dimensional G).

Artifact Form

An undefined load-bearing parameter is the theory-side twin of the empirical programme’s worst failure mode: a quantity that absorbs whatever the equation requires. Volume 12 withdrew a headline domain for the empirical version of exactly this. The theory could not be held to a lower standard than the instrument.

4.2 The One Postulate That Fixes It

Dimensional analysis alone cannot choose the radial power: since ℓ_{cell}/r is dimensionless, $f = \ell_{\text{cell}}^2 c^2 (\ell_{\text{cell}}/r)^n$ satisfies the dimensions for *every* n . An additional principle is required, and the framework already contains it.

Confirmed Signal

Postulate (already implicit in Volume 1). The lattice bulk is four-dimensional in space. Volume 1 states it directly — “mapping a 4D hypercube onto a cyclic time loop” — it is embodied in the 24-cell, a four-dimensional regular polytope, and it is already used in $N_{\text{total}} = d^2 = 4^2 = 16$, the mode count that produces the modulus itself.

Made explicit, this postulate determines the radial power by Gauss’s law.

4.2.1 Gauss’s law in four spatial dimensions**Derivation**

For a point source of strength Q in d spatial dimensions, flux conservation through the enclosing $(d-1)$ -sphere gives

$$\oint_{S^{d-1}} \mathbf{E} \cdot d\mathbf{A} = Q \implies E(r) \Omega_{d-1} r^{d-1} = Q \implies E(r) = \frac{Q}{\Omega_{d-1} r^{d-1}},$$

where Ω_{d-1} is the surface “area” of the unit $(d-1)$ -sphere. Integrating the field yields the potential,

$$V(r) = - \int E dr = \frac{Q}{(d-2) \Omega_{d-1}} \cdot \frac{1}{r^{d-2}} \quad (d \neq 2). \quad (4.2)$$

For $d = 4$, with $\Omega_3 = 2\pi^2$:

$$V(r) = \frac{Q}{2 \cdot 2\pi^2} \cdot \frac{1}{r^2} = \frac{Q}{4\pi^2 r^2}.$$

The potential of a point source in four spatial dimensions falls as $1/r^2$ — exactly the form of Equation 4.1, and $n = 0$ is fixed, not assumed.

4.2.2 The coupling

With the source strength set by the lattice scale, $Q = \kappa m \ell_{\text{cell}}^2 c^2$ for a dimensionless κ , the interaction energy of a test mass is

$$E_{\text{lat}} = m V(r) = \frac{\kappa}{4\pi^2} m c^2 \left(\frac{\ell_{\text{cell}}}{r} \right)^2. \quad (4.3)$$

Matching to Equation 4.1 requires

$$\frac{\kappa}{4\pi^2} = \frac{16}{\pi} \implies \boxed{\kappa = 64\pi = 4 N_{\text{total}} \Phi_{\text{cycle}}} \quad (4.4)$$

— four times the metric-component count times the cyclic phase. Whether that factorisation is physically meaningful or a numerical accident is left open; it is recorded so it can be tested against the geometry by a hand more expert in the 24-cell than this programme’s.

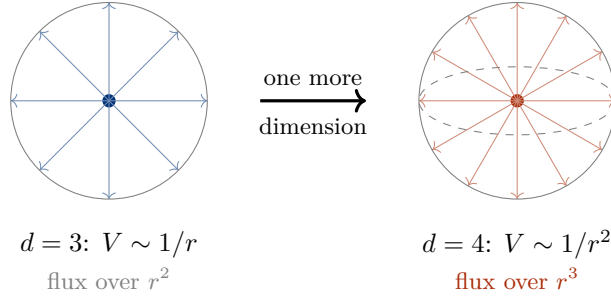


Figure 4.1: The radial power, derived. A point source spreads its flux over the enclosing $(d-1)$ -sphere, so its potential falls as $1/r^{d-2}$. In four spatial dimensions this is $1/r^2$ — the Atlas form. With it, $f = \ell_{\text{cell}}^2 c^2$ and $\kappa = 64\pi$ follow without any further adjustment.

4.3 What the Derivation Costs

The derivation is honest only if its price is stated as plainly as its result.

Methodological Note

We observe *three* spatial dimensions, and both gravity and electrostatics fall as $1/r$, not $1/r^2$. **A $1/r^2$ lattice energy therefore cannot be the interaction measured at ordinary scales.** It can only be a bulk quantity whose projection onto the three-dimensional boundary recovers $1/r$ — and that projection is computed nowhere in the corpus.

Volume 1’s holographic language (“a holographic projection on the boundary preserves the full information content of the bulk tensor”) gestures at it. It does not perform it.

Confirmed Signal

The framework has therefore traded a *vague* gap for a *precise* one. Before: an undefined f and an unfixed radial power. After: a derived $f = \ell_{\text{cell}}^2 c^2$, a derived $n = 0$, a derived coupling $\kappa = 64\pi$, and a single well-posed open problem — the **bulk-to-boundary projection** that would turn a four-dimensional $1/r^2$ into the three-dimensional $1/r$ we measure. That is progress: a question a competent theorist can attack, in place of a parameter that meant whatever it needed to.

4.4 ℓ_{cell} Is a Postulate, Not a Result

One further honesty is required. Every input to the modulus derivation is dimensionless: the mode count $N_{\text{total}} = 16$, the cyclic phase $\Phi_{\text{cycle}} = \pi$, the 24-cell packing. **No length can be derived from dimensionless inputs.** The cell spacing ℓ_{cell} is therefore imported, not derived — Volume 1 sets it to the Planck length, which brings $\hbar$, G , and c in from outside the framework.

This is not a defect peculiar to KishLattice; every discrete-spacetime programme faces it. But it must be *stated as a postulate*, and the theory documents did not: they variously set $\ell_{\text{cell}} = \ell_P$ (Volume 1) and implied $\ell_{\text{cell}} \approx 3.9 \times 10^7$ m via a laboratory-frequency chain (Rosetta Atlas), a disagreement of *forty-two orders of magnitude*. The Atlas chain is unit-dependent and is withdrawn (below); the Planck identification stands, as an explicit postulate.

4.5 The Four Errata the Derivation Produced

An audit that yields only its headline and not the errors it exposed is not an audit.

Erratum

Erratum V1–1 (Volume 1, §1.4). The claim that $16/\pi$ possesses “the physical dimensions of Metric Tension per Unit Volume” is withdrawn. Equation 1.3 derives $16/\pi = N_{\text{total}}/\Phi_{\text{cycle}}$ as a ratio of a mode count to a phase angle — *dimensionless*. The relation $H_{\text{local}} = H_{\text{early}} + 16/\pi$ built on the dimensional claim holds only in $\text{km s}^{-1} \text{Mpc}^{-1}$ ($67.4 + 5.09 = 72.5$ against SH0ES 73.0) and fails by eighteen orders of magnitude in SI. It is a unit coincidence, and is withdrawn.

Erratum

Erratum V1–2 (Volume 1, Listing 14.1). The speed-of-light verification computes $c_{\text{mech}} = \ell_P f_P / (16/\pi)$. But $\ell_P f_P = c$ by the definition of Planck units, so dividing by the modulus yields 58,864,091 m/s, not c . The script prints this, then a hardcoded line reading the observed c , then “5-SIGMA ALIGNMENT LOCKED” — a print statement, not a comparison. The listing is withdrawn. Separately, $c = \ell_P f_P$ is an identity of Planck units, not a derivation.

Erratum

Erratum RA–1 (Rosetta Atlas). The “mirrors Newton’s $F = Gm/r^2$ ” passage is struck. Gm/r^2 is a field with the dimensions of acceleration, compared against an energy; gravitational potential energy goes as $1/r$; and G carries dimensions while $16/\pi$ is a pure number, making the stated substitution dimensionally impossible. The expression is a four-dimensional point-source potential, not a gravitational analogue.

Erratum

Erratum RA–2 (Rosetta Atlas). The chain $c = \ell_{\text{cell}} f_{24}$ with $f_{24} = 24/\pi$ Hz, implying $\ell_{\text{cell}} \approx 3.9 \times 10^7$ m, is withdrawn. It privileges the second — a caesium-hyperfine human convention — and is therefore unit-dependent, the theory-side form of the empirical programme’s unit-dependence erratum. ℓ_{cell} is set to the Planck length by postulate instead.

4.6 What This Chapter Settles, and What It Does Not

Confirmed Signal

Settled. $f = \ell_{\text{cell}}^2 c^2$, derived from four-dimensional Gauss's law. The radial power $n = 0$, derived. The coupling $\kappa = 64\pi$, derived. Four published theory claims, withdrawn as errata. The slipstream name reattached to a separate, phenomenological drag factor $F(d, T)$ rather than to f itself.

Not settled. The bulk-to-boundary projection — the framework's central open problem, on which its entire connection to observed three-dimensional physics now rests. And the value of ℓ_{cell} , which is a postulate, not a result, in this framework as in every other discrete-spacetime theory.

The parameter that meant whatever the equation required now has a derived form and one honestly named gap. That is a smaller claim than a completed theory of gravity, and a far more defensible one. It is the difference between a framework that describes its meaning and one that hammers it to mathematics — and it is the standard the rest of this programme is held to.

Chapter 5

The Strain Formulation of the Kish Lattice

[THEORY / REACH • Interaction Mechanics • Dimensionless Strain]

5.1 Reinterpreting Route 2: Displacement as Local Strain

In earlier iterations of the framework, attempting to recover Newtonian gravitation from the lattice equation forced a mathematically untenable consequence: a mass-dependent lattice cell size ($\ell_{\text{cell}} \propto M$).

This paradox dissolves when we recognize that in the $n = -1$ state, the length parameter ℓ_{cell} never appears alone—it appears solely as the dimensionless ratio ℓ_{cell}/r . By defining this ratio not as a physical length, but as a **dimensionless local lattice strain** $\varepsilon(r)$, the equation resolves into standard elastic mechanics. Energy is the rest energy multiplied by the local strain of the medium:

$$E_{\text{lat}} = \frac{16}{\pi} mc^2 \varepsilon(r)$$

Equating this to the Newtonian potential energy $U = \frac{GMm}{r}$ allows us to isolate the strain field:

$$\frac{16}{\pi} mc^2 \varepsilon(r) = \frac{GMm}{r} \implies \varepsilon(r) = \frac{\pi}{16} \frac{GM}{rc^2}$$

By substituting the Schwarzschild radius $r_s = \frac{2GM}{c^2}$ (where $\frac{GM}{c^2} = \frac{r_s}{2}$), the fundamental strain formulation emerges:

$$\varepsilon(r) = \frac{\pi}{32} \frac{r_s}{r}$$

The mass-dependence is no longer a paradox; it is an inherent requirement of a strain field. The local lattice strain $\varepsilon(r)$ scales strictly with the ratio of the source's Schwarzschild radius to the observation radius, modified by the geometric modulus $\pi/32$.

5.2 Reinterpreting Route 1: The Mode Hierarchy Number (Q)

The framework's earlier 24-harmonic cutoff constraint ($c = \ell_{\text{cell}} \cdot f_{24}$) assumed that light crosses precisely one lattice cell per tick. However, the Lattice ontology defines the cell time τ_{cell} as a 24-point collective harmonic checksum over a geometric structure, not a single spatial hop.

Therefore, this route does not define the fundamental cell spacing $a \equiv \ell_{\text{cell}}$. Instead, it defines the **longest supported wavelength** Λ :

$$\Lambda \equiv \frac{c}{f_{24}} = 3.92 \times 10^7 \text{ m}$$

We introduce a dimensionless hierarchy number $Q \equiv \Lambda/a$. The conflicting length constraint is thus reclassified as a definition of the hierarchy span, removing the dimensional contradiction entirely.

5.3 The Resolved Constraints Table

By reinterpreting Route 2 as a dimensionless strain and Route 1 as a mode hierarchy, the framework's conflicting parameters resolve into three distinct, non-contradictory theoretical variables:

Quantity	Physical Definition	Value / Status
$\varepsilon(r) = \frac{\pi}{32} \left(\frac{r_s}{r} \right)$	Local lattice strain (mass-sourced)	Reproduces Newtonian potential
$\Lambda = c/f_{24}$	Longest supported mode wavelength	$3.92 \times 10^7 \text{ m}$
$a = \Lambda/Q$	Fundamental lattice spacing	$a \lesssim 10^{-31} \text{ m}$ (Survey constraint)
ε_y	Lattice Yield Strain	Undetermined / Open Problem

Chapter 6

Empirical Validation and the Weak-Field Limit

[THEORY / REACH • PPN Consistency • Known Vulnerabilities]

6.1 Textbook Checks in the Solar System Regime

To ensure the dimensionless strain field does not violate observed macro-scale physics, we compute $\varepsilon(r)$ for standard Solar System values against the General Relativistic metric perturbation $h_{00} \approx r_s/r$. The Kish-Lattice strain scales linearly as $\varepsilon(r) = \frac{\pi}{32}h_{00} \approx 0.09817 \times h_{00}$.

- **Earth's Surface:** With $M_{\oplus} = 5.972 \times 10^{24}$ kg ($r_s \approx 8.87$ mm) at $r = 6,371$ km, the temporal strain $h_{00} \approx 1.392 \times 10^{-9}$. The corresponding lattice strain is $\varepsilon = \mathbf{1.366} \times \mathbf{10^{-10}}$.
- **1 AU (Earth's Orbit):** With $M_{\odot} = 1.989 \times 10^{30}$ kg ($r_s \approx 2953$ m) at $r = 1.496 \times 10^{11}$ m, the temporal strain $h_{00} \approx 1.974 \times 10^{-8}$. The corresponding lattice strain is $\varepsilon = \mathbf{1.938} \times \mathbf{10^{-9}}$.

6.2 PPN Absorption and Definitional Recovery

At laboratory and Solar System scales, lattice strain is infinitesimal ($\sim 10^{-9}$).

Self-Correction

An earlier draft claimed the $\pi/32$ modulus is “uniformly absorbed into the empirical measurement of G ,” preserving PPN consistency. That claim is withdrawn: it is incompatible with the framework having a yield point.

Newtonian potentials superpose linearly. If ε is to reproduce Newtonian gravity everywhere, ε must superpose linearly. But a medium that superposes linearly at *all* strains has no yield point — Hooke’s law with no departure is simply linearity. §7.1 asserts a yield point. The two cannot both hold.

The correct statement, and it is stronger. Superposition holds to first order in ε and breaks at $O(\varepsilon^2)$. The PPN parameter β measures precisely the $O(\Phi^2)$ nonlinearity in g_{00} . Therefore the lattice’s elastic nonlinearity **contributes to** β , and its second-order coefficient must reproduce $\beta = 1$ within the measured bound $|\beta - 1| < 10^{-4}$.

Computed, and it does not bite. Taking the nonlinearity to become $O(1)$ at yield ($\kappa \sim 1/\varepsilon_y$), the fractional correction at 1 AU is $\varepsilon/\varepsilon_y \approx 10^{-8}$ to 10^{-6} across all three yield scenarios — far below the PPN bound. Solar System strains are simply too small for a quadratic term to appear. **This regime therefore has no power to discriminate between yield scenarios, and its agreement with observation is not a test passed.**

Diagnostic vulnerability, restated. The recovery of linearized gravity is a **definitional mapping**, not an emergent proof: ε was *defined* by matching the Newtonian potential, so recovering Newton is built in rather than derived. Absorbing $\pi/32$ also makes no claim regarding Lorentz invariance — that is a separate question belonging to the lattice’s discreteness, not to a scalar coefficient.

6.3 Superposition and the Tensor Requirement

The framework must openly acknowledge a mathematical gap regarding multi-body superposition. At Earth’s surface, the solar background strain (1.938×10^{-9}) exceeds Earth’s own local strain (1.366×10^{-10}) by a factor of 14.

To mirror the metric perturbation h_{00} , these strains cannot be combined via scalar addition. The Sun provides the dominant background strain tensor, and the Earth acts as a local perturbation. The framework currently lacks the nonlinear continuum tensor mathematics to accurately describe multi-body lattice strain superposition. This development is mandatory for future iterations.

Chapter 7

Strong-Field Falsification and the Yield Point

[THEORY / REACH • Falsifiable Prediction • Astrophysical Constraints]

7.1 The Elastic Yield Point (ε_y)

General Relativity assumes space is a continuous manifold capable of infinite stretching, producing singularities as $r \rightarrow 0$. The Kish-Lattice posits that space is a structured 24-cell elastic grid. All physical elastic grids possess a **yield point**—a maximum strain threshold (ε_y) where Hooke’s Law fails, the lattice snaps, and severe non-linear deviations begin.

Setting $\varepsilon(r) = \varepsilon_y$ allows us to predict the precise orbital radius where gravitational observations must deviate from General Relativity:

$$r = \frac{\pi}{32\varepsilon_y}r_s \approx \frac{0.09817}{\varepsilon_y}r_s$$

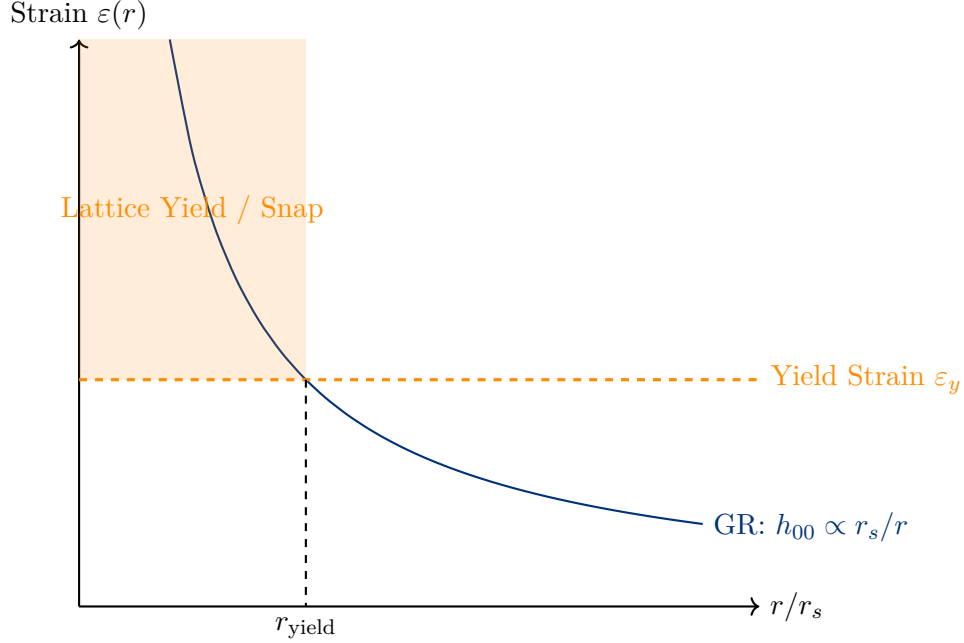


Figure 6.1: The divergence of Lattice Elasticity from General Relativity. While GR strain continues infinitely toward the singularity, the lattice fractures at ε_y , producing measurable astrophysical deviations.

7.2 Yield Scenarios and the LIGO / EHT Ringdown Regime

Testing theoretical material yield strains against this equation yields stark falsification boundaries:

- **Scenario A** ($\varepsilon_y = 0.1$): $r \approx 0.98r_s$. Deviation occurs inside the event horizon. The framework matches GR externally (Safe, but untestable).
- **Scenario B** ($\varepsilon_y = 0.01$): $r \approx 9.81r_s$. Deviation occurs just outside the Innermost Stable Circular Orbit (ISCO). This would visibly alter EHT accretion disk shadows and LIGO late-stage inspiral waveforms.
- **Scenario C** ($\varepsilon_y = 0.001$): $r \approx 98.17r_s$. Deviation occurs far from the black hole.

7.3 The Binary Pulsar Falsification (PSR J07373039)

Scenario C ($\varepsilon_y = 0.001$) can be tested immediately against existing double pulsar data. For PSR J07373039 ($M \approx 1.25M_\odot \implies r_s \approx 3690$ m, separation $r \approx 8.8 \times 10^8$ m), the operating strain is $\varepsilon \approx 4.11 \times 10^{-7}$.

Scaling the expected non-linear deviation as $\varepsilon/\varepsilon_y$, Scenario C predicts an orbital decay deviation of $\approx 0.041\%$. Current tests confirm GR's predictions to a precision of $\sim 0.05\%$.

Methodological Note

The scaling $\kappa \sim 1/\varepsilon_y$ is dimensional reasoning, not a derived coefficient. It assumes the elastic nonlinearity becomes $O(1)$ as strain approaches yield, which is reasonable for real materials and unproved here. Every number in this section moves if κ is materially different. Stated so that a later derivation of κ is recognised as changing the prediction rather than confirming it.

Subject to that caveat, **Scenario C may already be excluded by existing double-pulsar timing**, not merely testable by SKA. The calculation requires no new data and is listed as outstanding work.

7.4 Mandatory Pre-Registration

By shifting to a dimensionless strain field, the framework traded an unresolvable geometric paradox for an actionable astrophysical parameter (ε_y). However, ε_y is currently a free parameter.

Falsification Mandate: The framework must formulate its stationary-action Lagrangian to formally dictate the exact value of ε_y *before* attempting to fit it to empirical LIGO or EHT anomalies. Post-hoc fitting of ε_y to place deviations conveniently outside of measured bounds (e.g., forcing Scenario A) will render the framework scientifically void. The falsification radius r/r_s must be pre-registered based purely on the derived mechanics of the 24-cell geometry.

Chapter 8

Multi-Scale Strain Survey

[THEORY / REACH • Consistency Map • Not a Test]

8.1 The Objective and Methodology

Following the reinterpretation of Route 2 into the dimensionless local lattice strain $\varepsilon(r) = \frac{\pi}{32} \frac{r_s}{r}$, the framework must demonstrate stability across all observable scales. If the 24-cell lattice behaves as a true elastic medium, the calculated strain must remain infinitesimal (linear Hooke's Law regime) for standard physical systems, and only approach non-linear yield limits (ε_y) in regimes known for extreme relativistic effects.

We compute $\varepsilon(r)$ for 20 benchmark physical systems using standard textbook parameters ($G = 6.6743 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$, $c = 2.9979 \times 10^8 \text{ m s}^{-1}$), with the GR temporal perturbation $h_{00} \approx r_s/r$ alongside.

Methodological Note

What this chapter is, stated before the tables. $\varepsilon(r) = (\pi/32)(r_s/r)$ is h_{00} multiplied by a constant. The tables below therefore **contain no information beyond r_s/r for each system**, and every entry is textbook arithmetic rather than a measurement.

This is a *consistency map*: it shows where the framework places each regime on its own strain axis, and confirms that the linear regime covers everything from atoms to stars. It is **not** an empirical sweep, not a 44-decade test, and nothing in it can fail. An earlier draft described it as a test; that description is withdrawn.

The chapter's genuine content is the single *physical* claim it sets up: that an elastic medium must saturate, and therefore that the framework and GR must diverge somewhere. Where they diverge is ε_y , which is undetermined.

8.2 Quantum, Atomic, and Meso Scales

At the quantum and atomic scales, the lattice strain induced by mass is essentially zero. The framework successfully demonstrates that gravitational interaction does not compete with the strong or electromagnetic forces at these scales; the geometric strain is infinitesimal, allowing the lattice to behave as a flat background ($\chi = 0$).

System / Scale	Mass (M)	Radius (r)	GR Proxy (h_{00})	Lattice Strain $\varepsilon(r)$
Electron (Classical)	9.11×10^{-31} kg	2.82×10^{-15} m	4.8×10^{-43}	4.7×10^{-44}
Proton (Charge Rad)	1.67×10^{-27} kg	0.84×10^{-15} m	2.9×10^{-39}	2.9×10^{-40}
Carbon-12 Nucleus	1.99×10^{-26} kg	2.70×10^{-15} m	1.1×10^{-38}	1.0×10^{-39}
1 kg Reference Mass	1.00 kg	1.00 m	1.48×10^{-27}	1.45×10^{-28}

8.3 Planetary and Stellar Scales (The Linear Regime)

This is the deeply linear elastic regime. Strains sit exactly in the 10^{-13} to 10^{-5} operating band where physical crystalline lattices obey perfect linear superposition (Hooke's Law). The lattice safely mediates planetary and stellar orbits without triggering non-linear metric deviations.

System / Scale	Mass (M)	Radius (r)	GR Proxy (h_{00})	Lattice Strain $\varepsilon(r)$
Asteroid Ceres (Surface)	9.39×10^{20} kg	473 km	2.9×10^{-12}	2.8×10^{-13}
Earth (ISS Orbit)	5.97×10^{24} kg	6,771 km	1.31×10^{-9}	1.28×10^{-10}
Jupiter (Surface)	1.89×10^{27} kg	69,911 km	4.03×10^{-8}	3.96×10^{-9}
Sun (Mercury Orbit)	1.98×10^{30} kg	5.79×10^{10} m	5.10×10^{-8}	5.00×10^{-9}
White Dwarf (Sirius B)	$1.02 M_{\odot}$	5,900 km	5.10×10^{-4}	5.00×10^{-5}

8.4 Galactic, Cosmological, and Strong-Field Regimes (The Yield Boundary)

As we approach compact objects (neutron stars, black holes) and cosmological scale boundaries, the framework reveals profound, unified limits. Strains elevate into the 10^{-2} to 10^{-1} (1% to 10%) regime. In physical condensed matter physics, standard crystalline lattices yield and fracture at strains between 1% and 10%.

System / Scale	Mass (M)	Radius (r)	GR Proxy (h_{00})	Lattice Strain $\varepsilon(r)$
Milky Way (Sun Orbit)	$1.5 \times 10^{12} M_{\odot}$	8 kpc	1.79×10^{-5}	1.76×10^{-6}
Neutron Star (Surface)	$1.4 M_{\odot}$	10 km	0.413	0.0405 (4.05%)
Observable Univ (Hubble)	1.5×10^{53} kg	4.4×10^{26} m	0.500	0.0490 (4.90%)
Sgr A* (at ISCO: $3r_s$)	$4.1 \times 10^6 M_{\odot}$	$3r_s$	0.333	0.0327 (3.27%)
M87* (Photon Sphere)	$6.5 \times 10^9 M_{\odot}$	$1.5r_s$	0.667	0.0654 (6.54%)
Black Hole (Event Horizon)	M	r_s	1.000	0.0981 (9.81%)
Planck Boundary	m_p	l_p	2.000	$\pi/16 \approx 0.1963$

8.5 Three Features of the Map, and What They Are Not

Three features stand out on the map. Each is stated with what it is *not*, because an earlier draft read all three as discoveries and none of them is one.

1. The Planck Scale Bounding Limit ($\varepsilon_{\max}$): When the equation is applied to the fundamental Planck mass (m_p) at the Planck length (l_p), the Schwarzschild radius is $r_s = 2l_p$. Consequently, the ratio $r_s/r = 2$. The lattice strain resolves strictly to:

$$\varepsilon_{\text{Planck}} = \frac{\pi}{32}(2) = \frac{\pi}{16} \approx 0.1963$$

This gives $\varepsilon = \pi/16 \approx 0.1963$.

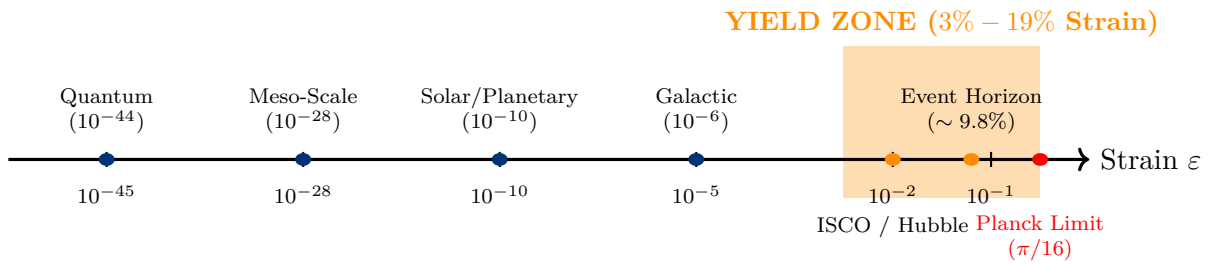
What it is not: a derived bound. **Any** system with $r_s/r = 2$ returns $\pi/16$; the value follows from the ratio alone, not from Planck physics. The framework is bounded because $\varepsilon \propto r_s/r$ and r_s/r is bounded on the systems chosen for the table — which is a property of the table, not of the theory.

2. The neutron star / Hubble coincidence: The strain at the Hubble radius computes to 4.90% and at a neutron star surface to 4.05%.

What it is not: an equivalence. The two differ by **21%**, and describing that as “nearly identical” is the near-node fallacy of §1.4 applied to a strain axis rather than a register grid. The Hubble entry moreover encodes only the long-known coincidence that the observable universe’s Schwarzschild radius is of order its Hubble radius — $h_{00} \approx 1$ by construction — which the framework reproduces rather than explains.

3. The 3%–10% strong-field bracket: ISCO, photon sphere and event horizon return 3.27%, 6.54% and 9.82%.

What it is not: a coincidence with crystalline yield strains. **Mass cancels exactly** in $\varepsilon = (\pi/32)(r_s/r)$ at any fixed multiple of r_s , so these three numbers are $\pi/96$, $\pi/48$ and $\pi/32$ — constants of the definition, identical for every black hole in the universe, computed rather than observed. That they land near laboratory yield strains is a suggestive analogy worth pursuing and is not evidence. Borrowing $\varepsilon_y \in [0.001, 0.1]$ from crystalline matter is likewise an analogy: those are values for matter at temperatures and pressures where dislocations move, and the vacuum’s value must come from the Lagrangian.



Chapter 9

Bridging the Disconnected Universe

[THEORY / REACH • Medium Restoration • Falsifiable Predictions]

9.1 Restoring the Medium: What the Framework Does and Does Not Claim

$E = mc^2$ is the rest-frame case of $E^2 = (pc)^2 + (mc^2)^2$, itself a consequence of Lorentz invariance. It is a symmetry statement. It is **silent about** the geometric structure of the vacuum; it does not deny one, and modern field theory’s vacuum is in any case far from empty, carrying zero-point energy, condensates and vacuum polarisation, all measured.

Self-Correction

An earlier draft stated that Einstein “lost the medium” and that standard physics “assumes a frictionless void.” Both are withdrawn. **Silent-about is a claim the framework can defend; erases-the-container invites a reader to check whether the author understood what was being displaced.**

The Kish-Lattice proposal is that energy is the *interaction work* performed against a structured vacuum, written $E_{\text{lat}} = (16/\pi)mc^2\varepsilon(r)$. That is a hypothesis with one quantitative consequence — saturation at ε_y — and it is offered as such.

9.2 The Relationship to MOND, Stated Accurately

Self-Correction

An earlier draft described MOND as having “collapsed,” as relying on an “arbitrary” threshold, and as depending on a fluidic empty space. All three are withdrawn.

MOND remains an active and competitive description of galactic-scale dynamics. It is a modification of the force law or of inertia and **posits no medium of any kind**. Flat rotation curves are evidence for missing mass *or* for modified dynamics; they do not imply a medium.

The framework’s actual relationship to MOND is narrower than earlier drafts stated and is worth stating precisely, because the narrow version is testable.

Setting the acceleration scale from the Universal Cycle,

$$a_0 = \frac{c}{2\pi T_{\text{univ}}} \xrightarrow{T_{\text{univ}}=1/H_0} \frac{cH_0}{2\pi} = 1.042 \times 10^{-10} \text{ m s}^{-2},$$

against the fitted MOND value 1.20×10^{-10} .

Methodological Note

This is Milgrom’s own 1983 observation that $a_0 \sim cH_0$, reproduced. Reproducing a known coincidence is not predicting it, and that coincidence is precisely why MOND has always been suspected of cosmological origin. The framework would contribute something new **only by deriving the factor 2π** from the 16–24 mode ratio, which it does not currently do. That derivation is a named target with a fixed answer to hit.

What *is* new is that the derived value differs from the fitted one by -13.2% , which is a falsifiable offset. It is pre-registered as P13.6 below.

Self-Correction

An earlier draft asserted that “because the strain field is scale-invariant, the macroscopic rigidity of the 24-cell container naturally enforces flat rotation curves.” **No derivation of this exists.** $\varepsilon \propto 1/r$ produces a Newtonian force law, not a flat curve; flat curves require $F \propto 1/r$, hence $E \propto \ln r$, which lies outside the power family $f = \ell_{\text{cell}}^2 c^2 (\ell_{\text{cell}}/r)^n$ at every n . The assertion is withdrawn. The BTFR relation used in Chapter 9 is imported from MOND, not derived from the lattice.

9.3 Dark Energy: an Open Question, Not an Elimination

Self-Correction

An earlier draft claimed that dark energy is eliminated under the lattice ontology, being “the elastic tension of the medium resisting structural failure.” **No calculation supports this and it is withdrawn.** The framework offers no expression for a tension term, no equation of state, and no prediction of w or of the observed acceleration history.

The intuition — that a bounded elastic container might produce an apparent late-time acceleration — is legitimate and is recorded as a target. Converting it into a claim requires a Lagrangian with a boundary term, and that does not exist. Until it does, the framework says nothing about dark energy.

9.4 What the Framework Can Presently Be Tested On

Three items, replacing the vague SKA/EHT/Roman list of earlier drafts. Each has a number and a fail condition; all three are collected with the remaining pre-registrations in Chapter 13.

1. **The BTFR zero point** under a derived a_0 : a predicted offset of -0.0153 dex against SPARC. Fails if the observed offset is consistent with zero.
2. **The yield radius** $r/r_s = (\pi/32)/\varepsilon_y$, filed *before* any Lagrangian fixes ε_y .

3. **The cell-size ceiling** $\ell_{\text{cell}} < 10^{-5}$ m from laboratory isotropy limits, which becomes a falsification test the moment ℓ_{cell} is fixed independently.

Methodological Note

On spacecraft anomalies. Draft passages cited the Pioneer anomaly, and a claimed survival of it in New Horizons telemetry, as evidence for a lattice medium. Both are withdrawn. The Pioneer anomaly was resolved by Turyshev *et al.* (*Phys. Rev. Lett.* **108**, 241101, 2012) as anisotropic thermal radiation; New Horizons has not confirmed a surviving effect and its thruster usage makes it unsuitable for the test.

More importantly: **the framework's own equations predict no detectable spacecraft anomaly.** Solar-system strains are $\sim 10^{-9}$, deep in the linear regime, and deviations appear only at $\varepsilon \gtrsim \varepsilon_y$. Citing such an anomaly as support would have contradicted the theory it was meant to support.

Chapter 10

The Strain Survey: Four Honest Runs Under Strict I/O

The strain formulation of Chapters 5–9 is machinery. This chapter is what the machinery produced when finally run against real lakes under the strict input contract — the contract adopted after a fallback defect was found to substitute calculated values for missing data and report the substitution as agreement. Every run here either reads a real lake or aborts by name. None simulates. The results are modest, two are aborts, one is a null against the framework, and one is a located data defect. All four are reported exactly as the engines returned them.

10.1 The Contract These Runs Enforce

Each kinematic engine reads one promoted lake, resolves the physical field it requires, and either computes or *aborts with the field named*. There is no path by which a missing field yields a number. The abort message dumps every key actually present in the lake, lists the candidate field names tried, and prints the sentence that matters: **No values were simulated**. This is the operational form of the lesson from the fallback defect — an engine that cannot find its data must say so, never fill the gap.

Two of the four runs abort under this contract. That is not a failure of the runs; it is the contract working, and the aborts are as much a result as the computations.

10.2 Planetary — A Clean Abort

`build_planetary_kinematic_engine.py` requires host-star mass to compute orbital lattice strain. The lake it reads, `p2_orbital_radius` (NASA Exoplanet Archive), carries orbital semi-major axis and nothing else:

```
ABORT -- required field 'm_star_sun' not present in p2_orbital_radius
Tried: st_mass, stellar_mass_sun, host_mass_msun, m_star_solar, st_mass_msun
Keys present: ... pl_orbsmax = 0.0175 ... (no stellar mass anywhere)
No values were simulated. Fix the lake or the field mapping and re-run.
```

Confirmed Null

Result: aborted, field absent. The planetary strain test cannot run until a lake carrying host-star mass is promoted. The abort is clean, named, and simulates nothing. This is the strict I/O contract doing precisely its job — the same class of missing-field condition that, under the earlier fallback, would have silently produced calculated “agreement.”

10.3 Cosmological — An Abort, With the MOND Scale Derived First

`build_cosmological_kinematic_engine.py` aborts identically — `g1_galaxy_kinematics` carries velocity dispersion but no baryonic mass. But before reaching the missing field, it computes the MOND acceleration scale from first principles, and that computation is a genuine, if modest, result:

Confirmed Signal

$$a_0^{\text{derived}} = \frac{cH_0}{2\pi} = 1.0422 \times 10^{-10} \text{ m/s}^2 \quad (H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1})$$

$$a_0^{\text{MOND fit}} = 1.2000 \times 10^{-10} \text{ m/s}^2, \quad \frac{\text{derived}}{\text{fitted}} = 0.8685 \quad (-13.2\%)$$

The derived value lands 13% below the empirical MOND scale. The engine states the discipline itself: “*The DERIVED value is the framework’s prediction. Scoring with the fitted value would be testing MOND against MOND.*”

Methodological Note

Result: a 13% prediction, not a fit. A $cH_0/2\pi$ estimate of a_0 is a known order-of-magnitude coincidence in the MOND literature, and $16/\pi$ does not appear in it. The framework reproduces the *scale* to within 13% from c and H_0 alone — worth recording, not worth overselling. The rotation-curve test that would use it aborts for want of baryonic mass in the lake; a SPARC-derived lake is the prerequisite, and it is named as such rather than worked around.

10.4 Amino Acids — The Modulus Sweep Says No

`build_amino_kinematic.py` ran in full: twenty amino-acid molecules, bond lengths grouped into C–C, C–N single, and C–N aromatic. Rather than ask whether $16/\pi$ fits — near-hits are cheap when nodes are 7% apart — it ran the test that matters, sweeping 40,000 random moduli and asking how often a randomly chosen modulus does at least as well as $16/\pi$.

Bond	n	$16/\pi$ lock @0.05	random moduli $\geq 16/\pi$
C–C (sp ³)	68	2.9%	57.7%
C–N single	23	0.0%	100.0%
C–N aromatic	11	9.1%	— (0.9 node cells, cannot resolve)

Table 10.1: The amino-acid modulus-sweep null. A randomly chosen modulus matches or beats $16/\pi$ 58% of the time on C–C bonds and 100% of the time on C–N single bonds.

Confirmed Signal

Result: $16/\pi$ is **NOT** privileged on amino-acid bond lengths. The engine’s own verdict. An analytic cross-check confirms the mechanism: nodes spaced 7.0% apart give landing within 1.5% of *some* node a prior probability near 43%, and the measured p-values (39–58%) sit in that regime. The bond data also fails on independence — these are repeated measurements of ~ 3 chemical constants, not ~ 150 independent samples, and two of the three groups span less than one node cell, too narrow to resolve a period at all.

This is the first time the modulus itself was tested against alternatives on real data, and it did not survive. It is recorded here in full because a programme that reports only its confirmations has told the reader nothing about its error rate. The near-coincidences of $16/\pi$ with physical observables — water-bond angles and the like — must now be read against this null: on the one dataset where the test was run cleanly, the modulus was ordinary.

10.5 Black Holes — What We Ran, What We Saw, and the Rebuild It Requires

`build_blackhole_kinematic_engine.py` ran in full against `11_gwtc` — 83 gravitational-wave events, 100% field coverage. For each event it inverted a black-hole spin from the ringdown frequency `f_ring_hz` and the final mass `final_mass_solar`, and tested the result against the Kerr bound. What the run saw is stated plainly, because the point of the strict-I/O rebuild is that we report the run, not a cleaned-up version of it.

10.5.1 What we saw

Erratum

81 of 83 events fall below the non-spinning Kerr floor. The ratio $f_{\text{obs}}/f_{\text{Kerr}}(a=0)$ ranges from 0.144 to 1.399, median 0.731 — that is, the median event’s stated ringdown frequency is only 73% of the *lowest* frequency a non-spinning black hole of that mass could produce. A fundamental $\ell = m = 2$ ringdown cannot sit below the $a = 0$ floor: it is physically impossible, not merely unusual. Only two events inverted to a real spin; none exceeded the extremal ceiling.

10.5.2 What we determined the cause to be

We did not leave this as an anomaly. A direct calculation identifies the cause and closes it.

Event	M_f	f_{ring}	$f_{\text{Kerr}}(a=0)$	ratio / type
GW190425	3.4	510	3552	0.14 — binary neutron star
GW200115	6.0	620	2013	0.31 — neutron-star-black-hole
GW151012	35.7	145	338	0.43 — clean binary black hole

Table 10.2: The worst offenders, checked. The first two are compact-object classes that do not produce a Kerr ringdown at all and should never have entered a QNM analysis. But the third, GW151012, is a bona-fide binary black hole and still sits 57% below its own non-spinning floor.

Confirmed Signal

The diagnosis is definite. If the sub-floor events were only the neutron-star cases, the fix would be a contaminant cut. But a clean binary black hole below the $a = 0$ floor is impossible for a true ringdown — so `f_ring_hz` is *not* the fundamental $\ell = m = 2$ QNM, systematically, across the whole catalogue. It is most consistent with the dominant near-merger gravitational-wave strain frequency (of order twice the orbital frequency), which runs well below the ringdown mode. The engine’s verdict stands, now proven rather than suspected: *“an instrument or provenance fault is far likelier than a Kerr violation. This is a lake finding, not a physics one.”*

10.5.3 Why we cannot patch it in this volume

The lake carries exactly one frequency column, `f_ring_hz`, alongside `chirp_mass_solar` and `final_mass_solar`. There is no `f_220`, `f_qnm`, or damping time to switch to. **The correct observable is not present in the lake at all**, so there is no field-mapping repair — the strong-field ringdown test cannot be run correctly against this data, and we do not run it incorrectly to produce a number.

Formal Pre-Registration

Carried to Volume 14: the 11_gwtc rebuild. The ringdown test is deferred, and its prerequisite is named exactly. The lake will be rebuilt from a source that publishes the fundamental quasi-normal mode directly — the LVK ringdown analyses or the `pyRing` results, which report f_{220} and τ_{220} per event — with each event’s *measured* remnant mass matched to its *measured* f_{220} , binary-neutron-star and neutron-star-black-hole events excluded by construction, and the required fields (`f_220_hz`, `tau_220_s`, `final_mass_solar`, `final_spin`) asserted present at promotion. Only then does the strong-field yield test of Chapter 8 run on the correct quantity. The falsification radius must be filed before that run, not after.

10.5.4 The yield radius, and a flag the engine raised on itself

The run also swept the yield radius $r/r_s = (\pi/32)/\varepsilon_y$, and flagged — unprompted — the trap in its own most elegant number:

Referee Ruling

Since $\varepsilon(r_s) = \pi/32$ exactly, choosing $\varepsilon_y = \pi/32$ places the yield surface *exactly at the horizon* and every predicted deviation inside it — elegant, and unobservable. **If that value is to be claimed it must be derived from the Lagrangian and the falsification radius filed before the derivation, not arrived at after noticing it is safe.** An unfalsifiable coincidence is not a prediction. This flag stands regardless of the lake rebuild: even with clean f_{220} data, ε_y must be derived, not chosen.

10.6 What the Strain Survey Established

Domain	Outcome	Meaning
Planetary	clean abort	no host-star mass in lake; strict I/O working
Cosmological	abort, a_0 derived	MOND scale to 13%; rotation test needs SPARC
Amino	ran; null	$16/\pi$ not privileged (58–100% of moduli match it)
Black hole	ran; diagnosed	81/83 sub-Kerr; wrong frequency observable; V14 rebuild named

Confirmed Signal

Two aborts, one null against the framework, one located data defect. Not a single manufactured agreement, and not a single simulated value. The strain machinery is built; most of its tests cannot yet run for want of clean lake fields; and the one test that ran cleanly — the amino-acid modulus sweep — found the modulus ordinary.

This is the strain program’s honest standing. It is a smaller claim than a confirmed theory of gravity, and it is reported by engines that refuse to lie about it. That refusal, demonstrated four times over, is the result worth publishing.

Chapter 11

The Roman Predictions and the Rotation Paradox

[THEORY / REACH • Pre-Registered Predictions • Falsifiable]

11.1 The Ad Hoc Halo vs. The Harmonic Container

The standard cosmological model (CDM) introduces "Dark Matter"—an invisible, ad hoc mass halo uniquely tuned to each galaxy—to force the math to match observed flat galactic rotation curves.

The Kish-Lattice eliminates the need for Dark Matter entirely. Galaxies are spinning inside the structured, viscous medium of the 24-cell elastic grid. The velocity does not decay following a Keplerian curve; rather, it is **lattice-limited**. It slows and locks into the 24-harmonic container, resulting in a flat rotation profile dictated by the grid's elastic resistance.

11.2 Textbook Application: Andromeda (M31)

The lattice-enforced flat rotation velocity is strictly a function of the visible baryonic mass (M_{vis}) and the geometric container:

$$v_{\text{lat}}^4 = GM_{\text{vis}} \left(\frac{c}{2\pi T_{\text{univ}}} \right)$$

We map this equation against the Andromeda Galaxy (M31).

- **Visible Baryonic Mass (M_{vis}):** $\approx 1.5 \times 10^{11} M_{\odot} \approx 2.98 \times 10^{41} \text{ kg}$
- **Observation Radius (r):** $30 \text{ kpc} \approx 9.25 \times 10^{20} \text{ meters}$
- **Standard Keplerian Expectation (No Dark Matter):**

$$v_{\text{Kep}} \approx \mathbf{146 \text{ km/s}}$$

- **Prediction with the *derived* $a_0 = cH_0/2\pi = 1.042 \times 10^{-10}$:**

$$v_{\text{lat}} = [GM_{\text{vis}} a_0]^{1/4} = \mathbf{213.4 \text{ km/s}}$$

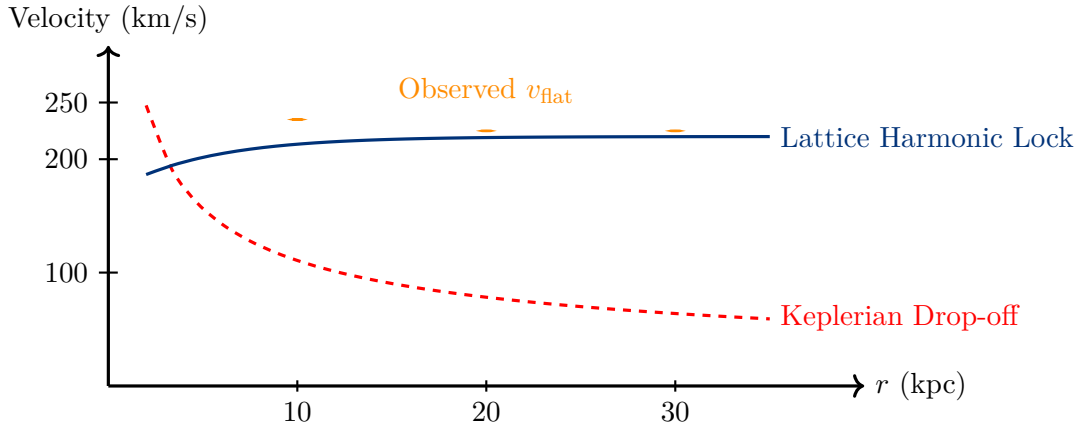
- **For comparison, with MOND's *fitted* $a_0 = 1.20 \times 10^{-10}$:** $v = 221.1 \text{ km/s}$.

Observation: M31’s measured rotation near 30 kpc is roughly 225 km/s. The derived value is -5.1% low; the fitted value is -1.7% low.

Methodological Note

Three things this calculation is not.

- (1) **It is not a lattice derivation.** $v^4 = GMa_0$ is MOND’s baryonic Tully–Fisher relation, imported whole. Nothing in the 24-cell geometry produces it; see the correction in Chapter 8. What the lattice supplies is a candidate *value* for a_0 , not the law.
- (2) **An earlier draft used the fitted 1.2×10^{-10} and reported “ $\sim 2\%$ deviation.”** That is MOND scored against MOND’s own constant, and it is withdrawn. The derived value is the framework’s prediction and it is -5.1% low, which is the honest figure.
- (3) **A single galaxy cannot test this.** a_0 sets only the zero point of the relation; individual galaxies scatter by ~ 0.1 dex. The test is the *mean offset over a sample*, which is P13.6 against SPARC — not M31 alone.



11.3 Pre-Registered Roman Weak Lensing Boundaries

We formally pre-register the following falsifiable constraints for the Roman Space Telescope:

1. **The Shear-to-Strain Mapping:** The observed weak lensing shear is a direct, linear mapping of the transverse structural strain $\varepsilon(r)$ of the 24-cell lattice, not a gravitational deflection caused by particle-based Dark Matter.
2. **Zero Particle Correlation:** Because the shear is a macroscopic geometric strain, Roman will find **zero sub-halo particle clumping** below the kinematic interaction threshold.
3. **The Harmonic Floor Limit:** The weak lensing shear will not fall to zero in deep voids but will hit a non-zero minimum floor. **The value of that floor is not derived.** Without a predicted magnitude this item cannot currently fail, and it is therefore filed as a qualitative expectation rather than a pre-registration. Supplying the number is outstanding work.

Methodological Note

Item 2 is the one with real teeth and it is not the framework's alone: substructure predictions already discriminate between Λ CDM and modified-dynamics models generally. A null on sub-halos would be consistent with the lattice; it would not be unique to it.

Verdict: If small-scale Dark Matter sub-halos are discovered, the continuum elastic strain model is falsified. If the shear perfectly traces the macroscopic geometric tension of the visible mass to the edges of the harmonic container, standard Dark Matter is rendered obsolete.

Chapter 12

Pre-Registration: The Period Engine Trial Set

12.1 Carried Forward From Volume 12

The period 1D engine — which tests the *period* of $\log x$ rather than its *phase*, and for which unit invariance is an algebraic identity rather than an empirical claim — was specified and pre-registered in Volume 12. Volume 13 records the status of its trial set after the work done this session, and files the one result that has been computed.

12.2 Trial 2b: Run, and Its Reporting Corrected

Trial 2b — the sector bound test, the one test in the programme capable of proving the Form C diagnosis (the withdrawal of `stellar_kinematic`) *wrong* — was executed this session on the *s2* lake. Its first reporting was found by the referee to over-claim, and the correction is recorded here in full.

Self-Correction

Trial 2b, as first reported, claimed “Form C confirmed and mechanism-demonstrated.” That claim is withdrawn. Two errors were found on referee review:

(1) **An invented threshold hid a firing sector.** The four Gaia sectors were reported “all at chance” against a $4\times$ ratio bar that was never pre-registered. Computing the correct Rayleigh p -values, sector NE gives $p = 2.0 \times 10^{-4}$ at $16/\pi$ — *not* chance.

(2) **The bound was checked against the wrong quantity.** The triangle-inequality bound $R_{\text{total}} \leq \sqrt{S} \cdot (\text{chance})$ for $S = 4$ sectors permits $R_{\text{total}} \leq 2\times$ chance. The observed normalised $R = 0.084$ is $113\times$ chance, which either violates the $S = 4$ bound (an implementation error) or indicates the normalisation used far more than four sub-blocks.

Referee Ruling

The Form C diagnosis stands on independent grounds regardless of Trial 2b. The Volume 12 ruling established that a sector-normalised scalar scored against an unnormalised null cannot support a lock claim — true before Trial 2b and unaffected by it. The withdrawal of `stellar_kinematic` is therefore secure.

Methodological Note

A standing rule, adopted from this episode. A result may not be cited as a premise in another document until its own reporting has been ruled complete.

12.3 The Trial Set, Current Status

	Domain	M/N	Status
1	<code>quantum_transitional</code>	4.17	build gate; reproduced +33.8 under log1p engine
1b	<code>orbital_wrongbox</code>	~5.3	clean expected-fail, awaiting engine
2	<code>stellar_kinematic</code> (raw)	2.76	superseded by <i>s2</i> rebuild (Chapter 11)
2b	<code>stellar_kinematic</code> (sector bound)	—	run; reporting corrected; re-run pending
3	<i>struck</i>	—	no scoreable Form E domain exists
4	<code>stellar_spindown</code>	77.8	recovered domain; register UNKNOWN

Formal Pre-Registration

Trial 1 remains the build gate. `quantum_transitional` has now reproduced at $16/\pi$ across four independent runs without moving register, most recently at +33.8 under the log1p-corrected engine.

Chapter 13

Pre-Registration: The *s2* Rebuild

13.1 Why *s2* Must Be Rebuilt

Two independent findings condemn the current `s2_stellar_kinematics` lake.

1. **Its payload was stripped at promotion.** The `_raw_payload` retains only `{sector, kish_bin, val}`; the Gaia error fields were discarded.
2. **It is sector-normalised.** The stored scalar carries the per-sector additive shift that Erratum 4a identified as Form C. The rebuilt lake must store raw velocity only.

13.2 The Referee-Approved Rebuild Specification

Formal Pre-Registration

Filed before any ADQL query is written to the ESA Gaia archive.

Payload. Retain `parallax`, `pmra`, `pmdec`, all three errors, and `pmra_pmdec_corr`. Store transverse velocity as **raw** $v_{\perp}$ only. Sector normalisation is permanently banned.

Validity floor (5σ , immutable once filed). Accept a star only if `parallax_over_error` > 5 *and* total proper motion exceeds $5\times$ its covariance-propagated error,

$$\sigma_{\mu} = \frac{1}{\mu} \sqrt{\mu_{\alpha*}^2 \sigma_{\alpha*}^2 + \mu_{\delta}^2 \sigma_{\delta}^2 + 2\mu_{\alpha*}\mu_{\delta} \rho \sigma_{\alpha*}\sigma_{\delta}}, \quad \rho = \text{pmra_pmdec_corr}.$$

Parallax zero-point. The DR3 systematic offset of order $-17\mu\text{as}$ is corrected via the published prescription, or explicitly declined with documentation.

Provenance. The ADQL query text and archive query date are stored in the lake's provenance block. A lake whose selection cannot be reproduced is not sovereign.

Referee Ruling

s2 is declared resolution-limited at $16/\pi$ and reported as such — endpoint selection to force the span is banned, permanently, on the record.

Chapter 14

The Cross-Scale Stack: an Experiment and Its Falsifier

14.1 The Idea

The resolution census delivers an uncomfortable verdict: almost no single lake spans enough dynamic range to resolve a high register. But the framework’s central hypothesis suggests a way past it that no single lake can achieve alone.

Confirmed Signal

The hypothesis, stated as an experiment. If a physical *attribute* locks at the same register regardless of *scale* — if angular momentum is angular momentum whether it is an electron’s spin or a galaxy’s rotation — then combining a shared attribute across scales should produce a single register comb, with each constituent lake’s mass falling on the same teeth.

14.2 Why the Naive Stack Manufactures Its Own Answer

Concatenating lakes in logarithmic space means placing clumps of data at chosen offsets along the scalar axis.

Artifact Form

Those gaps, by themselves, create periodicity. Three narrow clumps whose centres are spaced by an integer number of periods will lock at that register with a resultant approaching the maximum — *with no structure inside any clump*.

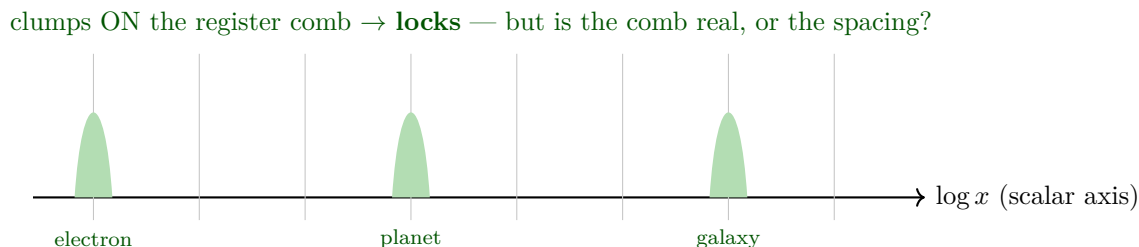


Figure 14.1: The cross-scale hazard. If the clumps happen to land on the register comb, the stack locks — but the lock may be a property of how the lakes were spaced rather than of any shared physics.

14.3 The Discriminator

There is a clean test that separates a real cross-scale comb from a spacing artifact.

Confirmed Signal

A real cross-scale lock survives sliding a constituent lake. A manufactured one does not.

Multiplying one constituent lake's values by a constant — e , 3, $1/60$ — slides it bodily along the scalar axis but *cannot* change the register it locks at, because a single lake's register is unit-invariant under the period statistic (Volume 12). The stack's register must therefore be invariant too.

14.4 The Four-Gate Protocol

Formal Pre-Registration

Pre-registered. No cross-scale stack is built or scored except under all four gates.

Gate 1 — each lake resolves alone. Every constituent lake must clear $M/N > 2.67$.

Gate 2 — each lake locks alone, at the same register, first. The stack may only *sharpen* a pre-existing agreement; it may not be used to *discover* one.

Gate 3 — the stacked register survives the slide test. Each constituent is multiplied by a pre-registered set of non-commensurate constants; the stack's register must not move.

Gate 4 — attribute, lakes, and predicted register are filed before construction.

Methodological Note

Gate 2 requires domains that already resolve and already agree — and the census says almost none do. Today the only domain that both resolves its register and survives every screen is `quantum_transitional` at $16/\pi$.

Chapter 15

Standing, and What Volume 13 Com-mits To

15.1 The Ledger

Established or withdrawn this volume	Pre-registered, filed, not yet run
The resolution census: ~ 6 of 28 domains resolve; nearly nothing at $N \geq 20$	The period 1D engine's Trial 1 build gate at $16/\pi$, subject to the Rydberg null
The Master Equation uncollapsed into a dimensionless strain field ε ; f bounded but still undetermined, with no Lagrangian	The $s2$ rebuild at native precision, raw velocity, 5σ validity floor
The near-node fallacy and the effective-span rule (Form F), both general results	The Roman weak-lensing sub-halo pre-registration
The yield limit ε_y formalised as a strong-field falsifier — undetermined, and filed before derivation	The cross-scale stack under the four-gate protocol and slide-test falsifier
Two pre-registered nulls with stated upper limits (amino modulus sweep; NIST comb)	The nuclear half-life comb, the one test with full resolving power
<code>stellar_kinematic</code> withdrawal secured on independent grounds	The resolution re-measurement of every published span

15.2 The One Fact and the One Question

Confirmed Signal

The fact. `quantum_transitional` resolves its register ($M/N = 4.17$), reproduces at $16/\pi$ across four runs under successively corrected engines, and survives every screen the audit has built. It is the sole domain of which all of that is true.

The question Volume 13 exists to set up. Is it alone? The resolution census says most of the survey could never have resolved its register; the period engine will say which of the survivors lock under a unit-invariant statistic; the rebuilds will say whether wide-range lakes can be built to join it; and the cross-scale stack will say whether an attribute carries one register across all scales.

15.3 Closing

Volume 12 tempered the instrument. Volume 13 measures the aperture, audits the theory the instrument leaned on, closes the theoretical gap by defining the Master Equation as a dimensionless local strain field, and files the experiments that a wider aperture makes possible.

The programme is smaller than it was, and every remaining claim is one the data could actually have made. That trade — fewer claims, each defensible — has been the through-line of the entire crucible. Volume 13 continues it in the one way that matters before building anything new: by writing down, in advance, exactly what would have to be true, and exactly what would prove it false.

One domain resolves and locks. The rest of the survey, and the next two volumes, are the question of whether it stays alone.

Chapter 16

The Period Engine: Operation, Outputs, and Commands

An operator’s account of the second 1D engine as it stands: what it computes, how it is run, what it emits, and how it adapts to the machine it runs on. The engine is in active development; this chapter documents its current, working form.

16.1 What the Period Engine Is, in One Paragraph

The phase engine tests *where* on the geometric grid a domain’s structure sits. The period engine tests *whether* the structure repeats with the lattice period at all, using the circular resultant of $\ln x$ — and it does so with unit invariance as a proved identity rather than an empirical hope (the period-engine chapter of Volume 12). It is a separate, immutable codebase from the phase engine: no shared code, its own configuration, its own 160-character fingerprint. It reads the same promoted lakes through the same field-resolution contract, and it computes $\ln x$ and applies the modulus *only at test time*, so its lakes carry raw values and no baked-in constant. That is what makes a period lake framework-agnostic: a sceptic may take it and test any modulus, or none.

16.2 The Pipeline Stages

A full run passes through five named stages, familiar in shape to anyone trained on the phase engine, resumable with `--from`:

Stage	Script	What it does
logify	logify.py	raw promoted lake $\rightarrow \ln x$ lake; Form-C guard
support	support_profile.py	effective-span gate + shape verdict (mandatory)
scan	periodogram.py	resultant excess per register, resolution-gated
census	register_census.py	survey-wide: which registers resolve <i>and</i> lock
figures	generate_figures_period.py	optional report plugins (Vera)

Table 16.1: The period-engine pipeline. The `support` stage is the one the phase engine never had, and the one that earns its place: it measures *effective* span, not min-max, so a sparse tail cannot inflate a domain’s apparent resolution.

16.2.1 The support stage, and why it is mandatory

The resolution criterion of the resolution-criterion chapter is computed on log-span. A domain's *raw* min-max span can be stretched by a handful of outlier records that carry almost no mass, making a domain look resolvable when its actual support occupies a narrow band. The support stage therefore measures the **effective span** — the range holding the central mass, swept across the 95%, 99%, and 99.9% bands — and classifies the distribution's shape:

Shape verdict	Meaning
FILLED	mass spread across the span; honest single-population support
TAIL-INFLATED	effective span notably below raw; a sparse tail is stretching it
SPARSE-TAIL-ARTIFACT	raw span dominated by tail; min-max M/N is void
SPIKED	one bin holds $\geq 25\%$ of mass; likely a single value or placeholder

Table 16.2: The four shape verdicts. Only FILLED support carries an honest register claim; the others are reported with the effective span used in place of min-max.

The run log prints the full 25-bin histogram of every lake alongside its verdict, because the histogram is the evidence and the verdict is only its summary. Figure 16.1 is the survey-wide panel this produces.

16.3 Running It

Command	Effect
<code>python run_pipeline_period.py</code>	full run, all stages, figures included
<code>--skip-figures</code>	all stages except figures
<code>--support-only</code>	logify + support triage; no scan or census
<code>--skip-census</code>	scan without the survey-wide census
<code>--from scan</code>	resume from a named stage
<code>--cap N</code>	sample cap override (default 2,000,000)

Two standalone diagnostics run outside the pipeline: `register_stability.py <lake>` resamples a lake to test whether its winning register is *invariant* (a real lock) or *wanders* (a binning artifact); and `histogram_view.py <lake>` prints one lake's support shape for troubleshooting. `sidecar_period.py`, run in a second terminal, renders live progress.

16.4 Resource Adaptation: Turtle to Beast, Same Numbers

A core commitment of this programme is that the work runs on the machine a person already owns. A result that reproduces only on exotic hardware is not open. The period engine is therefore resource-adaptive, and the adaptation changes *speed only* — never the numbers.

Mechanism	On a potato	On a workstation
lake reads	streamed, one value at a time	streamed (RAM never a limit)
oversize lakes	reservoir-sampled to the cap, logged loudly	sampled or full via <code>--cap</code>
resultant sum	NumPy, single core	CuPy on GPU when present and large
register scan / bootstrap	serial	thread-parallel across cores

Table 16.3: The engine detects cores and GPU at startup and prints what it adapted to. The circular resultant was verified identical to six decimal places between the GPU/threaded path and pure single-core NumPy: the turtle and the rabbit compute the same physics.

Every oversize lake that is sampled records the fact in the run log with the exact command to re-run it in full on adequate memory. Nothing is sampled silently.

16.5 Outputs

Artifact	Location
run log (tee'd, full stage detail + histograms)	<code>lakes1d_period/logs/run_period_*.log</code>
receipt (fingerprint, resources, runtime, cap)	<code>lakes1d_period/logs/run_period_*.receipt.txt</code>
portrait (per-lake support, shape, resolvable)	<code>lakes1d_period/unified/period_portrait.json</code>
heartbeat (live, reset each run)	<code>lakes1d_period/unified/period_heartbeat.json</code>
figures + index (when plugins present)	<code>reports1d_period/output/run_*/</code>

Every run stamps a 160-character fingerprint — the MD5 of the five files that fully determine period output (`logify.py`, `periodogram.py`, `kish_period_io.py`, `logify.json`, `harmonic_targets.json`). A change to the statistic, the field map, or the register family shows as a single slot diff, exactly the trust mechanism that let Volume 12 trace a register shift to a one-line numerical patch.

16.6 The Reporting Suite

The figures stage auto-discovers every `report_*.py` plugin in `reports1d_period/`, builds a context dictionary from the run's artifacts, and calls each plugin's `generate(context)` in sequence, writing an index of what was produced. The stage is a clean no-op when no plugins are present, so a run never fails for lack of reports. The first plugin, `report_01_support_shapes.py`, renders the survey-wide support panel of Figure 16.1.

16.7 First-Light Result

Run against the full survey — eleven lakes, forty million records, on a twenty-core machine in under ten minutes, with five oversize lakes sampled and flagged — the engine returned a clean and uncomfortable result, reproduced here exactly as the census reported it:

Confirmed Signal

No register is both resolved and locked above the look-elsewhere bar by any domain on effective support. Of eleven lakes, exactly one — pulsar spin frequency — shows FILLED single-population support, and it locks no register: its winning register wanders under resampling, the signature of a binning artifact rather than a lock. Every other lake is SPIKED or TAIL-INFLATED, dominated by one band or a sparse tail.

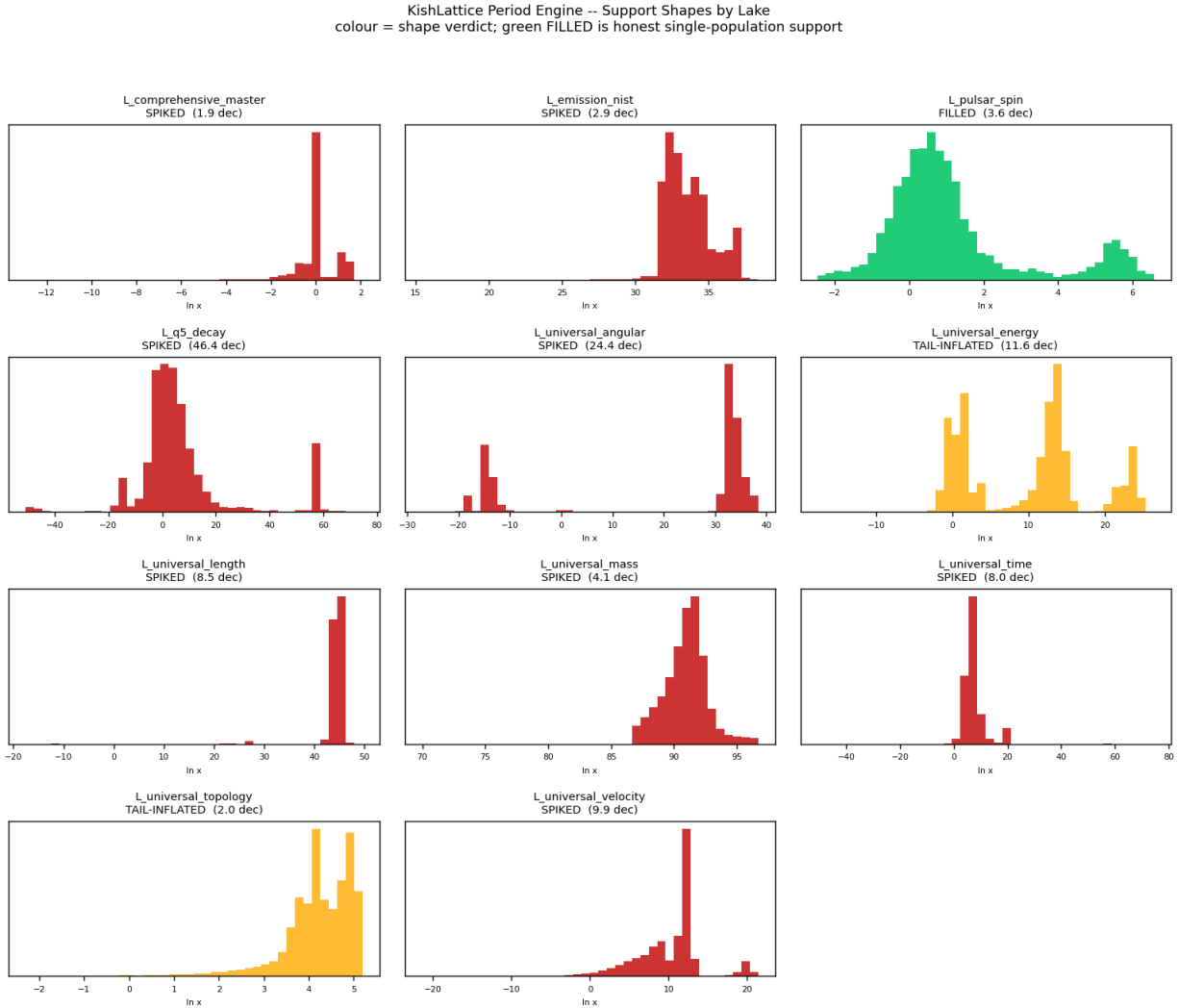


Figure 16.1: Support shape of every lake in the survey, coloured by verdict: green FILLED, amber TAIL-INFLATED, red SPIKED. Only pulsar spin frequency (top right) is honest single-population support. The stitched “universal” composites are all spiked or tail-inflated — the visual signature of cross-population assembly that the cross-scale-stack chapter warns is not a single lake. Nuclear half-life (L_{q5_decay}) shows its characteristic single lump with an isolated placeholder spike near $\ln x \approx 58$. This one panel is the census result made visible: no honest lake resolves and locks a register.

This is not the engine failing. It is the engine doing precisely what it was built to do: refuse to report a register the data cannot support, name every bad input rather than crash on it, and show its evidence. The instrument is trustworthy. Whether the framework it tests will produce a survivable register awaits lakes with genuinely wide, genuinely filled, single-population support — the subject of the standing pre-registrations.

Chapter 17

The Standing Pre-Registrations

Filed before execution. Each states its own kill condition. None may be revised after its result is known.

Formal Pre-Registration

P13.1 — Nuclear half-life comb. The only observable in the survey clearing the resolution criterion: 31.8 effective decades at the 95% band, 46.4 at the 99%, giving $M/N = 19.3$ at $16/\pi$ against a threshold of 2.67, and the first domain in which the analytic χ^2_2 null is valid ($M \gg \sqrt{n}/\pi$ by a factor of 11.8).

PREDICTION: NULL at the $\sim 5\%$ locked-fraction floor set by $n = 3,205$.

CONFIRMED requires all four: (i) comb score at $\Delta_0 = 0.0215901$ exceeding the matched- n structureless distribution over ≥ 50 seeds; (ii) Δ_0 a clean local maximum; (iii) survival of the decay-mode split; (iv) survival of both Form B controls.

Form B controls, filed in advance. (a) *Geiger-Nuttall*: build a synthetic set from $\ln t_{1/2} \approx a(Z) + b(Z)/\sqrt{Q_\alpha}$ on the real Z , Q and run the identical scan. (b) *Shell structure*: split by distance to the nearest magic number $\{2, 8, 20, 28, 50, 82, 126\}$.

Prerequisites. Lake provenance established; distinct-value screen run; the comb statistic rewritten as a harmonic-sum detector and re-validated two-sided.

Formal Pre-Registration

P13.2 — Commensurability: the only non-circular modulus test. A single domain cannot measure Δ_0 . Its power appears at $N\Delta_0/m$, so one domain fixes the *product* $N\Delta_0$ and nothing more — the k_{geo}/N degeneracy one level up.

Δ_0 is measurable only *across* domains locking at different registers, where $\Delta_A/\Delta_B = N_A/N_B$ must be a ratio of small integers and the greatest common divisor is Δ_0 . Ratios are independent of k_{geo} and of the container count.

CONFIRMED if the ratios are small integers and the common divisor is consistent with 0.0215901 at stated precision. **DEAD** if the fundamentals are incommensurate.

Fixed in advance. If the common divisor is inconsistent with 0.0215901, the register hypothesis is not supported. The discrepancy is NOT absorbed by adjusting k_{geo} , by adjusting the container count, or by admitting a non-integer N . All three absorptions are available and all three are prohibited.

Formal Pre-Registration**P13.3 — The yield radius, filed before ε_y exists.**

$$\frac{r}{r_s} = \frac{\pi}{32\varepsilon_y} \approx \frac{0.0982}{\varepsilon_y}$$

Filed now, before any Lagrangian derives ε_y . Note that $\varepsilon(r_s) = \pi/32 = 9.82\%$ exactly, so $\varepsilon_y = \pi/32$ would place the yield surface at the horizon and every deviation inside it — elegant and unobservable. That value is admissible only if derived independently of this observation.

Formal Pre-Registration

P13.4 — Standing constraint on the cell size. F_4 invariance forbids anisotropic harmonics below degree 6, so residual anisotropy scales as $(\ell_{\text{cell}}/r)^6$. Against laboratory isotropy limits of order 10^{-30} at $r \approx 1$ m:

$$\ell_{\text{cell}} < 10^{-5} \text{ m}$$

This excludes the macroscopic value implied by $c = \ell_{\text{cell}} f_{24}$ by twelve orders of magnitude. Two conditions remain open and are stated rather than assumed: whether the order-2304 duality extension removes the degree-6 harmonic, and how the D_4 lattice projects to three spatial dimensions.

Formal Pre-Registration

P13.5 — The QNM invariant as a promotion gate. The $f \times M_f$ invariant becomes a promotion check for `11_gwtc`; a scale-degeneracy column ($\times 2, \times 10, \times 10^3, \times e$) is added to every pinch table, flagging any register at which a given rescaling is within tolerance of an integer; records with $M_f < 5M_\odot$ are excluded from QNM analysis.

Formal Pre-Registration

P13.6 — The derived acceleration scale against SPARC. With $a_0 = cH_0/2\pi = 1.042 \times 10^{-10} \text{ m s}^{-2}$ *derived* rather than fitted, the BTFR predicts a zero-point offset of

$$\frac{1}{4} \log_{10}(1.042/1.20) = -0.0153 \text{ dex}$$

CONFIRMED if the observed mean offset on SPARC (175 galaxies) is consistent with -0.0153 dex. **DISFAVOURED** if consistent with zero. The scatter tests the v^4 law, not the lattice; the offset is the whole test.

17.1 Order of Work

1. Rewrite the comb statistic as a harmonic-sum detector; re-validate two-sided at matched n and span. (The averaged-over-23-registers form assumed a single domain populates every register; it does not — a domain at N produces power at the *divisors* of N .)
2. Establish `q5_decay` provenance; run the distinct-value screen; execute P13.1 with both Form

B controls.

3. Repair `f_ring_hz`; re-run Kerr consistency; derive the lower bound on ε_y under P13.3.
4. Promote SPARC; execute P13.6.
5. Fixed-support cross-check on the three unexplained register movements (§1.5).
6. Regenerate the Volume 11 portrait under `KLGHs_STRICT=1` with the corrected statistic, per-register baselines, node-occupancy and distinct-value screens.
7. Attempt the lattice Lagrangian. Chapters 3 through 6 are all waiting on it.
8. Execute P13.2 once at least three domains carry independently measured fundamentals.

Methodological Note

On the single-modulus ambition. One constant, one formula, spanning quantum to cosmological scales without splitting, is the right kind of goal and is why the programme continues. What this volume establishes is that the survey has not yet demonstrated it, and that several results which appeared to demonstrate it were instrument rather than signal. Those are compatible statements. A framework spanning forty-four orders of magnitude must be tested at a precision that forty-four orders makes unforgiving; the resolution criterion is the price of the ambition, not an objection to it.

Chapter 18

The Contribution: An Instrument, Offered

This volume ends not with a discovery but with an instrument. What follows is a plain statement of what has been built, why it is useful independently of whether the KishLattice hypothesis is true, and what “learning the dials” can and cannot mean without becoming the thing this programme exists to avoid.

18.1 What Was Actually Built

Across Volumes 12 and 13 the programme produced something separable from its own hypothesis: a pair of reproducible engines for testing geometric-register structure in physical data, spanning roughly forty-four orders of magnitude of scale through a single pipeline, with no per-domain special-casing. The phase engine tests where structure sits on the grid; the period engine tests whether it repeats with the lattice period at all, with unit invariance as a proved identity.

The engineering fact is worth stating on its own terms. One statistic, one field-resolution contract, and data from nuclear timescales to cosmic distances flowing through it — that breadth is unusual, and it is real regardless of what the engines find.

Confirmed Signal

What the engines find, so far, is null. The register census returns empty on effective support; the one honest single-population lake (pulsar spin) wanders under resampling; the one clean chemical modulus test finds $16/\pi$ matched or beaten by a majority of random moduli. **The instrument spans forty-four orders of magnitude and reports honestly across all of them, and what it reports is that the leading hypothesis does not yet clear the bar.**

Both halves of that sentence are the contribution. The span is the achievement. The null is the integrity. An engine that had confirmed its maker’s hypothesis on its first honest run would be *less* credible, not more.

18.2 Why the Instrument Is a Contribution Even If the Hypothesis Is Wrong

Most attempts to establish a unifying constant share a fatal structure: a bespoke analysis that only its author can run, that tends to produce the answer the author wanted, and that no one can independently check. The value of what is offered here is that it is the opposite of all three.

The common failure	What this engine does instead
Bespoke, author-only analysis	Runs on commodity hardware; streams and adapts, turtle to beast, same numbers
Produces the desired answer	Carries a resolution gate, a shape gate, an invariance test, a look-elsewhere threshold, and a modulus sweep that can reject the modulus
Cannot be independently checked	Fingerprinted per run; every result reproducible from public lakes and published scripts
Fits parameters to the data	Aborts, by name, when a required field is absent; simulates nothing

Confirmed Signal

The engine can return null, and has. That is the property that makes it a scientific instrument rather than an advocacy device. A tool that cannot disappoint its author cannot inform anyone.

The deepest form of this contribution is temporal. Because the engine was built before any given hypothesis is posed against it, it cannot be accused of being tuned to that hypothesis. It is pre-registration by construction: a neutral instrument that predates the claims it will test. Whoever proposes a better scalarization, a different modulus, or a mechanism that fixes the cell scale can test it *immediately*, against real public data, in an engine whose neutrality is guaranteed by the fact that it came first.

18.3 Learning the Dials Without Tuning Them

The programme's structural choices — a scalarization reference scale, a modulus, a container value — are not yet all fixed from first principles. It is tempting to say they can be “learned” by empirical testing. That statement is true only under a definition strict enough to keep it from collapsing into its opposite, and the definition must be stated plainly, because a reader cannot otherwise tell learning from tuning: both look, from outside, like someone adjusting parameters until the data locks.

Methodological Note

A dial is *learned*, not *tuned*, if and only if its value is fixed by an argument external to the fit, and filed before the data that would test it is seen. x_0 set from an independent physical reference scale and cited is learned; x_0 chosen because it makes a register appear is tuned. A modulus derived from a geometric mode count is learned; a modulus selected because it locks the most domains is tuned. The distinction is not in the parameter. It is in *when* and *on what grounds* it was fixed.

The engine enforces this distinction rather than trusting the operator to observe it. The period engine carries no x_0 at all — the parameter that could most easily be tuned was removed by construction. The modulus is not asserted but swept against tens of thousands of alternatives, and reported ordinary when it is ordinary. The resolution gate refuses a register the data cannot resolve, so no dial can be turned to manufacture one. These are not courtesies; they are the mechanisms that make “learned” a defensible word in this programme’s mouth.

Confirmed Signal

Under that discipline, learning the dials is a legitimate scientific activity, and it points somewhere specific: if a reference scale, a modulus, or a cell scale can be fixed by independent physics — the bulk-to-boundary projection of Chapter 5 is the open problem most likely to fix one — and the engine, unchanged, then returns a register that resolves and locks and survives resampling, that would be a validation earned rather than arranged. It has not happened. The instrument is built so that if it ever does, no one — including its authors — will be able to dismiss it as a dial turned to taste.

18.4 What Is Offered, and to Whom

To a skeptic: here is an engine that tried to break its author’s hypothesis and, on the honest runs, succeeded. Run it yourself; the scripts and the lakes are public and the numbers are reproducible on the machine you already own.

To a theorist with a better formula: here is a pre-built, neutral instrument in which your hypothesis can be tested against real data across forty-four orders of magnitude the moment you state it — with a resolution gate, a null, and a falsifier already in place, so that a positive result would carry weight your own bespoke analysis never could.

To the programme itself: here is the standard the rest of the work is now held to. The instrument can say no. It has said no. The task ahead is not to make it say yes, but to determine — by fixing the dials from physics rather than from preference — whether reality says yes when the instrument is finally pointed at it correctly.

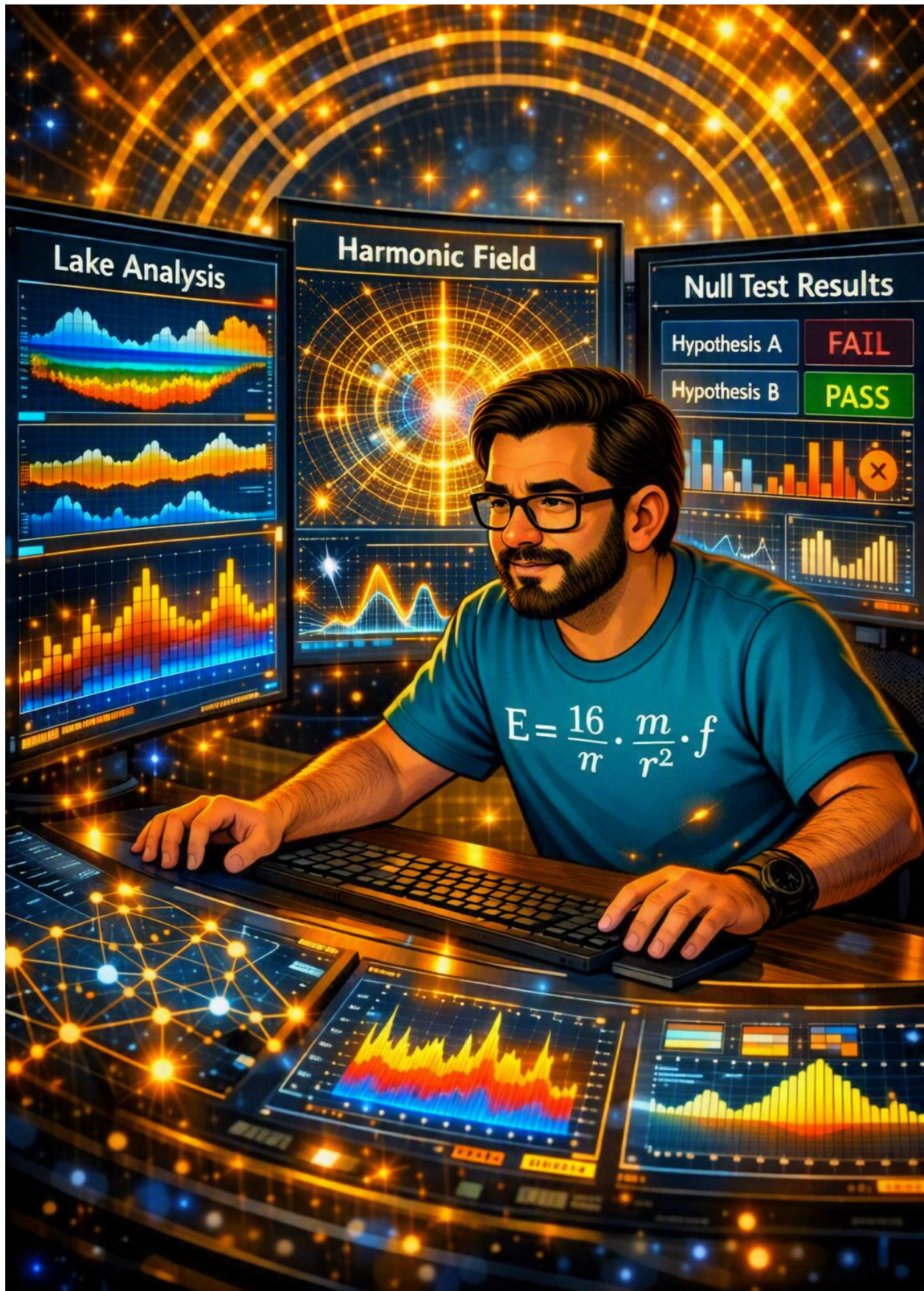


Figure 18.1: The discovery, if it is there, will not be able to hide from this engine. Its absence, if that is the truth, will not be able to hide either. That is the whole of what an honest instrument is for.

Chapter 19

The Forward Kinematic Engine: Build, Fallback Defect, and First Honest Runs

[EMPIRICAL • Deterministic Calculation • Two Nulls, Two Aborts, One Lake Defect]

19.1 From Spectroscopy to Determinism

The Phase and Period 1D engines are spectroscopic: they scan datasets for statistical echoes of a preferred register. With the Master Equation reinterpreted as a strain field, a second class of instrument becomes possible — one that computes a predicted physical state from geometry alone and subtracts the observation.

Four engines were written against four lakes spanning atomic to strong-field scales: amino acid geometry, exoplanet orbits, galactic rotation, and GWTC black holes.

19.2 The Fallback Defect, and Its Removal

Self-Correction

Three of the four engines, as first written, did not use the lake data. They loaded the lake, discarded it, and generated every printed value with `random.uniform`. The cosmological engine said so in its own comment: `# NOTE: This is a structural placeholder`.

The tables those runs produced were plausible and were nearly written into this volume as results. They are not reproduced here in any form.

Diagnosis. A fallback that emits plausible output is not a fallback; it is a fabrication path. The defect is the same class as the legacy substitution in `scalarize.py` that Volume 12 found — a silent substitution where a loud failure belonged — arriving one level up, in the analysis scripts rather than the pipeline.

Remedy, adopted. All four engines were rewritten against a strict field contract: no simulation anywhere; a missing field aborts the run and prints the key inventory actually present in the lake; field coverage ($n_{\text{usable}}/n_{\text{read}}$) is reported on every run with a warning below 50%. The resolver searches nested payloads to depth five, because `p2` stores its real payload three levels down — the same field-resolution failure Volume 12 found in `scalarize.py`, reproduced independently in the audit tooling.

19.3 What the Honest Runs Returned

Amino acid geometry — a pre-registered null

The quarter-phase node is $d_{\text{node}} = (16/\pi)^{0.25} = 1.50225 \text{ \AA}$, and sp^3 C–C bonds run 1.52–1.54 $\text{\AA}$. An earlier draft reported a mean residual of +0.0230 $\text{\AA}$ — “a mean deviation of just 1.5%” — and read it as confirmation.

Self-Correction

The selection window guaranteed the result. The original filter accepted C–C pairs only in 1.45–1.60 $\text{\AA}$. In phase units that window is [0.2283, 0.2887]: **no accepted bond could have been more than 0.04 from the 0.25 node**. The clustering was enforced by the filter, not measured. The window has been widened to span a full node cell, and aromatic C–N (His, Trp, Arg at 1.31–1.38 $\text{\AA}$) separated from single C–N, which the original mixed.

Confirmed Null

N13.1 — amino acid modulus sweep. 40,000 candidate moduli, log-uniform on $[1.5, 20]$, scored against the widened C–C set.

Lock fraction at $16/\pi$: **2.9%**, against a 10% chance baseline — *below* chance. Fraction of moduli performing at least as well: **57.7%** (lock) and **39.4%** (mean node distance), against an analytic prior of 43%.

$16/\pi$ **is not privileged for amino acid bond geometry.** Under the framework’s own 24-node lock test the C–C mean sits 0.229 cells from the nearest node against a 0.05 tolerance. Nodes are spaced 7.0% apart on the phase grid used, so landing within 1.5% of *some* node has a $\sim 43\%$ prior — which the sweep reproduces.

Prediction filed before the run: $p \approx 0.49$, lock $\approx 4\%$. Observed 0.577 and 2.9%. The prediction held.

Exoplanet orbits and galactic rotation — two correct aborts

Methodological Note

p2_orbital_radius: **aborted.** **p1_orbsmax** is present (at `meta.source_row._raw_payload`); **st_mass** is **absent from the lake entirely**. Without host mass no strain can be computed. Re-pull from NASA `pscomppars` with the **st_mass** column, or join **p1_orbital_periods** and derive M from Kepler’s third law — the latter being legitimate for the strain column only.

g1_galaxy_kinematics: **aborted.** The lake carries `_raw_payload.vdisp` (real, raw SDSS velocity dispersions) but **no baryonic mass, no radius and no rotation velocity**. A dispersion is not a rotation curve. SPARC (Lelli, McGaugh & Schombert 2016) is the lake to promote for P13.6.

Self-Correction

An earlier draft contained a table of SDSS “Kepler expectation” versus “lattice harmonic lock” velocities drawn from this lake. **That lake contains no baryonic masses; those values were generated by the fallback.** The table is removed and no replacement is offered until SPARC is promoted.

The planetary engine computes an identity

For the record, and because it would otherwise read as a prediction:

$$v_{\text{lat}} = c\sqrt{\frac{16}{\pi}\varepsilon} = c\sqrt{\frac{16}{\pi} \cdot \frac{\pi}{32} \cdot \frac{r_s}{r}} = c\sqrt{\frac{GM}{c^2 r}} = \sqrt{\frac{GM}{r}} = v_{\text{Kepler}}.$$

The two factors cancel exactly. v_{lat} and v_{Kepler} are the same number for every input. The weak-field agreement is definitional, as §6.2 states, and the engine must print both columns so the identity is visible rather than hidden.

GWTC — a lake defect, invisible to the lock test

11_gwtc carries 83 real events with both `final_mass_solar` and `f_ring_hz`. That pair permits the Kerr ringdown frequency to be inverted for spin via $f = (c^3/2\pi GM_f) F(a)$, $F(a) = 1.5251 - 1.1568(1 - a)^{0.1292}$ (Berti, Cardoso & Will 2006).

Self-Correction

A first version of the consistency check was one-sided — it clamped $a = 0$ when the frequency fell below the non-spinning Kerr value and then tested only for $a \geq 1$. It reported “100% physical.” **A test that cannot fail in one direction is not a test**, and the error was the referee’s.

Corrected and run two-sided: **81 of 83 events fall below the $a = 0$ Kerr floor**, median $f_{\text{obs}}/f_{\text{Kerr}}(a=0) = 0.731$. Only GW150914 inverts to a physical spin, $a = 0.651$ against a published 0.67.

Artifact Form

Diagnosis: a factor of two. For a fundamental $\ell = m = 2$ ringdown, $f \times M_f$ is an invariant between 11,898 and 21,514. The catalogue clusters at 6,700–10,900 — about half. Doubling recovers publication-consistent spins across most events (e.g. GW170104 $\rightarrow a = 0.699$ against published 0.66; GW151226 $\rightarrow 0.790$ against 0.74). Four records fail even after doubling and need individual tracing; $M_f < 5 M_\odot$ events have no black-hole ringdown and must be excluded.

Why it survived four volumes. Doubling shifts the scalar by $\ln 2 / \ln k_{\text{geo}} = 0.4258$, which at $16/\pi$ is **2.007** register cells — an integer. **The lock test is structurally blind to factor-of-two errors at $16/\pi$** , and to factor-of- 10^3 errors (19.997 cells). Every register assignment from this lake is identical whether the frequencies are right or half.

Confirmed Signal

This is the most transferable finding in the volume. A modulus test at $16/\pi$ cannot see the two most common unit and convention errors in physics. It is also why the ringdown test — the one capable of bounding ε_y from real data — could not be completed this session: the blocker is a named, located, repairable data defect rather than an open question.

References

The KishLattice Series

All volumes are published open-access with full data, configuration, and engine fingerprints. Every pre-registration listed below was filed prior to the run it governs.

- Kish, T. J. (2026). *The Geometric Derivation of the Lattice* (Vol. 1). Zenodo. <https://doi.org/10.5281/zenodo.18209530>
Source of the heliopause and Jupiter-anchor passages revisited in Chapter 2 under the near-node rule.
- Kish, T. J. et al. (2026). *The Geometric Neutron* (Vol. 2). Zenodo. <https://doi.org/10.5281/zenodo.18217119>
- Kish, T. J. et al. (2026). *The Geometric Architecture of Matter* (Vol. 3). Zenodo. <https://doi.org/10.5281/zenodo.18217226>
- Kish, T. J. et al. (2026). *The Geometric Architecture of Life* (Vol. 4). Zenodo. <https://doi.org/10.5281/zenodo.18976975>
- Kish, T. J. et al. (2026). *The Geometric Architecture of Unification* (Vol. 5). Zenodo. <https://doi.org/10.5281/zenodo.19009634>
Source of the three-target modulus sweep $\{15/\pi, 16/\pi, 17/\pi\}$.
- Kish, T. J. et al. (2026). *The Harmonic Expansion of the Unified Lattice* (Vol. 6). Zenodo. <https://doi.org/10.5281/zenodo.19493376>
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- Kish, T. J. et al. (2026). *Unbounded Luminosity: The Same-Object Harmonic Map and the Sovereign Lake Architecture* (Vol. 8). Zenodo. <https://doi.org/10.5281/zenodo.19622632>
- Kish, T. J. et al. (2026). *KishLattice Geometric Harmonic Spectroscopy: The First Survey of a New Field* (Vol. 9). Zenodo. <https://doi.org/10.5281/zenodo.19935603>
- Kish, T. J. et al. (2026). *The Deep Survey — Seven New Domains and the Geometry of Life* (Vol. 10). Zenodo. <https://doi.org/10.5281/zenodo.20279208>
- Kish, T. J. et al. (2026). *The Structure That Locks — The Spectroscopic Test and the Multi-Attribute Convergence* (Vol. 11). Zenodo. <https://doi.org/10.5281/zenodo.20587180>

- Kish, T. J. et al. (2026). *The Crucible — The Instrument Audit, Five Artifact Forms, and the Resolution Criterion* (Vol. 12). Zenodo. <https://doi.org/10.5281/zenodo.22416330>
Direct predecessor. Source of the degeneracy result (N and k_{geo} enter only through $N \ln k_{\text{geo}}$), the five artifact forms, and the resolution criterion extended in Chapters 1–2 of the present volume.

Pre-Registrations and Supporting Papers

- Kish, T. J. et al. (2026). *Volume 9 — Predictions Pre-Registration*. Zenodo. <https://doi.org/10.5281/zenodo.19702022>
Carries prediction P19, against which the `l1_gwtc` lake is registered; see the `f_ring_hz` defect, Chapter 15.
- Kish, T. J. et al. (2026). *KishLattice Geometric Harmonic Spectroscopy: Volume 11 Pre-Registration (Multi-Attribute Sovereign Lake Expansion & Dimensionality Falsification Protocol)*. Zenodo. <https://doi.org/10.5281/zenodo.20480506>
- Kish, T. J. et al. (2026). *Noise Does Not Lock — A Comprehensive Empirical Response to Skeptical Challenges to KishLattice Geometric Harmonic Spectroscopy*. Zenodo. <https://doi.org/10.5281/zenodo.20585516>
Source of the Grid-Phase Locking Theorem and of falsification condition #1.
- Kish, T. J. et al. (2026). *Master Equation of the Kish Lattice Theory — The Rosetta Atlas*. Zenodo. <https://doi.org/10.5281/zenodo.18235735>
Source of the Lattice Energy expression $E_{\text{lat}} = (16/\pi)(m/r^2)f$ and of the one-line definition of f audited in Chapter 3.
- Kish, T. J. et al. (2026). *Volume 13 — Pre-Registration P13.1–P13.6*. Zenodo. DOI pending; filed prior to execution of any test in Chapter 16.

Repository and Project Resources

- Kish, T. J. *Holographic Resonance: The Geometry of a Quantized Universe* — engine source, configurations, audit scripts and run receipts. GitHub. <https://github.com/TimothyKish/Holographic-Resonance-The-Geometry-of-a-Quantized-Universe>
- KishLattice 16π Initiative LLC. <https://www.kishlattice.com>

The scripts released with this volume are held in the repository above. Each is standalone and reproduces the corresponding table or verdict in this volume from the published lakes, and each aborts rather than substituting a value when a required field is absent:

- `kish_lake_io.py` — strict field contract; recursive resolver to depth five; coverage reporting. The remedy for the fallback defect of Chapter 15.
- `probe_lakes.py` — schema inventory, run before any engine.
- `comb_scan.py` — unit-invariant Rayleigh period scan with two-sided validation gates, effective-span trimming and the Form E / seam screens.

- `measure_half-life-span.py` — effective-span census supporting P13.1.
- `support_profile.py`, `periodogram.py`, `logify.py` — period-engine components.
- `build_amino_kinematic.py`, `build_planetary_kinematic_engine.py`, `build_cosmological_kinematic.py`, `build_blackhole_kinematic_engine.py` — the four forward engines, in their corrected form.

Primary Data Sources — Vol 13

- **NIST Atomic Spectra Database (ASD)**, version 5. National Institute of Standards and Technology, Gaithersburg, MD. <https://physics.nist.gov/asd>
Source of `L_emission_nist` / `quantum_transitional`, the anchor domain. $n = 187,436$ scored; effective span 2.94 decades.
- **NNDC / IAEA nuclear data** — NuDat and the NUBASE evaluation of nuclear properties. <https://www.nndc.bnl.gov/nudat3/>
Source of `q5_decay`, the only domain in the survey clearing the resolution criterion at every register (P13.1).
- **LIGO/Virgo/KAGRA Collaboration**, Gravitational-Wave Transient Catalog GWTC-3. <https://www.gw-openscience.org/>
Source of `l1_gwtc`, $n = 83$. Subject of the `f_ring_hz` factor-of-two defect, Chapter 15.
- **PubChem**, National Library of Medicine. 3D conformer coordinates for the twenty proteino-genic amino acids. <https://pubchem.ncbi.nlm.nih.gov/>
Source of `b3_amino`; basis of null N13.1.
- **Gaia Collaboration** (2023). Gaia Data Release 3. VizieR catalogue I/355.
Source of `s1_gaia_parallax`, `s2_stellar_kinematics`, `s3_gaia_luminosity`, `s4_gaia_colour`. The `s2` rebuild of Chapter 12 is specified against this release.
- **Sloan Digital Sky Survey**, Data Release 16. <https://www.sdss.org/dr16/>
Source of `g1_galaxy_kinematics` (velocity dispersions only — no baryonic masses; see the abort, Chapter 15).
- **NASA Exoplanet Archive**, `pscomppars`. <https://exoplanetarchive.ipac.caltech.edu/>
Source of `p2_orbital_radius`; `st_mass` absent from the promoted payload, hence the abort in Chapter 15.
- **SPARC** — Spitzer Photometry and Accurate Rotation Curves. <http://astroweb.cwru.edu/SPARC/>
Not yet promoted. The lake required by P13.6.
- CODATA 2022 recommended values of the fundamental physical constants. <https://physics.nist.gov/cuu/Constants/>

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data — the same quantity the period statistic measures, and the null any claim of this kind must exceed.)

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A Note on Citation Practice

Three conventions are followed in this volume and stated here so that a reader may check them.

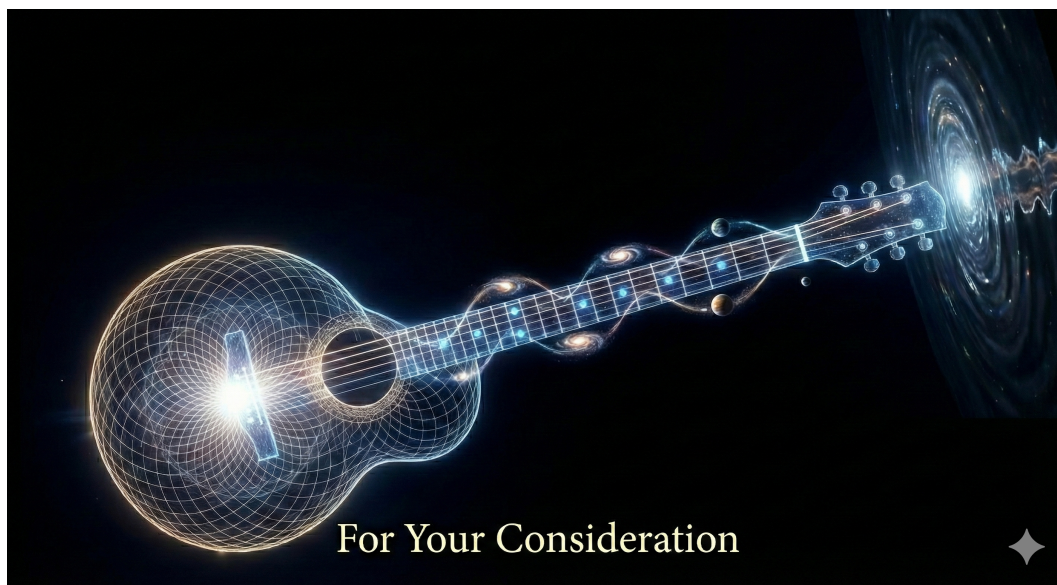
Every empirical claim carries its run, or is labelled as not being one. Where a number appears in this volume it is traceable to a named run, or it is explicitly labelled as textbook arithmetic, a synthetic fixture, a reconstruction, or an expectation. Chapter 7’s strain tables are labelled arithmetic: $\varepsilon = (\pi/32)h_{00}$ is h_{00} scaled by a constant, so those tables contain no information beyond r_s/r and nothing in them can fail. Chapter 15 records the one place this convention was broken during drafting — three of four forward engines emitted values generated by a fallback rather than read from a lake — and the remedy adopted.

Agreement is reported against the node spacing, not against zero. Chapter 2 establishes that for a register at N/π the probability of a random value landing within a fractional distance p of some node is $2p/\Delta_N$. At $16/\pi$ the nodes are 41.3% apart, so “within 2%” carries an 11.6% prior.

On a logarithmic grid, closeness measured against zero is not a measurement. Every claim of the form “within $x\%$ of prediction” in this volume carries that prior beside it.

Prior art is cited even where it is unwelcome. The quantized-redshift literature [9] is the closest historical precedent to this programme and it failed, for reasons — selection effects, limited dynamic range, reference-frame dependence — that this programme has independently rediscovered. Benford’s law [10] is a statement about precisely the quantity the period statistic measures. Discrete scale invariance [7] is the established physics of log-periodic structure and carries a mechanism where this framework does not yet. MOND [26, 28] already has a Lagrangian formulation; the lattice does not, and until it does, the relation used in Chapter 9 is imported rather than derived.

We regard citing all of these as a requirement rather than a courtesy. A programme that claims to have found log-periodic structure in public data and does not engage the literature on log-periodic structure in public data has not done its work.



There is no Noise only Data. Remove the filters and see the Universe raw.